HAIMAN'S CONJECTURE AND SPRINGER'S REPRESENTATIONS

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Abstract. For any element z of a finite Weyl group W, we give an explicit formula for the graded W-character of the intersection cohomology of a generic closed Lusztig variety for z, generalizing Abreu–Nigro's for the type-A case. We then study the Laurent polynomials describing how our formula decomposes into certain reducible characters determined by the Springer correspondence. In type BC, we conjecture that when these multiplicity polynomials are averaged over certain families studied by Lusztig, the nonzero coefficients are positive and unimodal. In any irreducible type, we conjecture that there is a Coxeter element c such that the same properties appear as long as z is c-sortable. This generalizes conjectures of Haiman and Shareshian–Wachs from symmetric groups to Weyl groups, resolving a 1993 question of Haiman. We also express the multiplicities in terms of unipotent Lusztig varieties, which yields some new formulas for Kazhdan–Lusztig polynomials.

1. Conjectures and Results

1.1. **Motivation.** The aim of this paper is to generalize conjectures of Haiman and Shareshian–Wachs from symmetric groups to Weyl groups.

For any finite Coxeter group W, let H_W be the Hecke algebra of W over $\mathbf{Z}[\mathsf{v}^{\pm 1}]$ with standard basis $\{\sigma_w\}_{w\in W}$. We fix a system of simple reflections $S \subseteq W$ and normalize the elements σ_w so that H_W, as an algebra, is generated by the set $\{\sigma_s\}_{s\in S}$ modulo the relations $\sigma_s - \sigma_s^{-1} = \mathsf{v} - \mathsf{v}^{-1}$ and so-called braid relations. Let $\{c_w\}_{w\in W}$ be the Kazhdan–Lusztig basis of H_W denoted $\{C'_w\}_{w\in W}$ in [KL79]. It is invariant under the ring anti-automorphism sending $\mathsf{v} \mapsto \mathsf{v}^{-1}$ and $\sigma_s \mapsto \sigma_s^{-1}$ for all $s \in S$. For instance, if $s \in S$, then $c_s = \mathsf{v}^{-1} + \sigma_s = \mathsf{v} + \sigma_s^{-1}$.

Let $\mathrm{Irr}(W)$ be the set of irreducible characters of W. The representation theory of H_W deforms that of W, just as H_W itself deforms the group ring $\mathbf{Z}W$. In particular, each class function χ on W deforms to a $\mathbf{Z}[\mathsf{v}^{\pm 1}]$-linear function χ_{v} on H_W that remains a trace: *i.e.*, vanishes on commutators. If W is a Weyl group and $\chi \in \mathrm{Irr}(W)$, then χ takes values in $\mathbf{Z}$, and χ_{v} takes values in $\mathbf{Z}[\mathsf{v}^{\pm 1}]$.

When W is the symmetric group S_n, the theory of cell modules shows that $\chi_{\mathsf{v}}(c_z) \in \mathbf{Z}_{\geq 0}[\mathsf{v}^{\pm 1}]$ for all $z \in W$ and $\chi \in \mathrm{Irr}(W)$, a positivity statement that already fails when W is the Weyl group of type BC_2. Haiman conjectured a much stronger positivity. To state it, first recall that the elements of $\mathrm{Irr}(S_n)$ are indexed by integer partitions of n. For any $\lambda \vdash n$, let $\chi_\lambda \in \mathrm{Irr}(S_n)$ be the corresponding irreducible character. Next, recall that the Frobenius characteristic map [Mac95, §1.7] is a linear

bijection between class functions on S_n and symmetric functions of degree n (say, in n variables), taking χ_λ to the Schur function indexed by λ.

For any $\mu \vdash n$, let $mon_\mu \colon S_n \to \mathbf{Z}$ be the class function whose Frobenius characteristic is the *monomial* symmetric function indexed by μ. Then

$$\chi_\lambda = \sum_{\mu \vdash n} K_{\lambda,\mu} mon_\mu,$$

where $K_{\lambda,\mu} \in \mathbf{Z}_{\geq 0}$ is the Kostka number counting semistandard Young tableaux of shape λ and weight μ. For any $z \in S_n$, let

$$\alpha_z^\mu(\mathsf{v}) = mon_{\mu,\mathsf{v}}(c_z) \in \mathbf{Z}[\mathsf{v}^{\pm 1}].$$

What follows is Conjecture 2.1 in [Hai93].

Conjecture A (Haiman)**.** *For any $z \in S_n$ and $\mu \vdash n$, the nonzero coefficients of $\alpha_z^\mu(\mathsf{v})$ are all positive and form a unimodal sequence.*

Haiman points out that the coefficients of $\alpha_z^\mu(\mathsf{v})$ are at least palindromic, due to the invariance of c_z under the anti-automorphism sending σ_s, v to $\sigma_s^{-1}, \mathsf{v}^{-1}$.

Further motivation for Conjecture A appeared in [CHSS16], where Clearman, Hyatt, Shelton, and Skandera showed that the positivity of $mon_{\mu,\mathsf{v}}(c_z)$ would subsume the Shareshian–Wachs conjecture: the e-positivity of the chromatic quasisymmetric functions of unit interval graphs [SW12]. Thus it would also subsume the Stanley–Stembridge conjecture [Sta95, SS93] recently proved in [Hik25, GMR+25].

To explain the connection, let $ind_\mu \colon S_n \to \mathbf{Z}$ be the class function whose Frobenius characteristic is the homogeneous symmetric function indexed by μ. Equivalently, ind_μ is the character of the representation of S_n induced from the parabolic subgroup $S_\mu \simeq S_{\mu_1} \times S_{\mu_2} \times \ldots$ Then

$$ind_\mu = \sum_{\lambda \vdash n} K_{\lambda,\mu} \chi_\lambda.$$

So the Laurent polynomials $\alpha_z^\mu(\mathsf{v})$ are uniquely determined by requiring that

$$\sum_{\lambda \vdash n} \chi_{\lambda,\mathsf{v}}(c_z) \chi_\lambda = \sum_{\mu \vdash n} \alpha_z^\mu(\mathsf{v}) ind_\mu \tag{1.1}$$

as functions on S_n.

The authors of [CHSS16] show that if z is 312-*avoiding*[1], meaning no indices $1 \leq i < j < k \leq n$ exist for which $z(j) < z(k) < z(i)$, then the Frobenius characteristic of the left-hand side of (1.1) is the transform, under Macdonald's involution ω, of the chromatic quasisymmetric function of a unit interval graph corresponding to z. Thus Conjecture A would imply the h-positivity of the ω-transform, equivalent to the e-positivity of the Shareshian–Wachs function.

For further details about unit interval graphs and their chromatic quasisymmetric functions, we defer to the references above.

[1] In Haiman's terminology, *codominant*.

1.2. **Lusztig Varieties.** At the end of [Hai93, §2], Haiman writes:

> There is probably an analog of Conjecture 2.1 for other Coxeter groups or at least Weyl groups. At present, however, it is not apparent what the analog of a monomial character should be.

Though we still lack a meaningful generalization of the characters mon_μ beyond S_n, we have found a way to generalize Conjecture A to irreducible finite Weyl groups, using the work of Springer and Lusztig in algebraic geometry.

Already in [SW12], Shareshian–Wachs had proposed an algebro-geometric formula[2] for the chromatic quasisymmetric function of any unit interval graph, in terms of an S_n-action on the singular cohomology of a complex algebraic variety associated with the graph. In Lie-theoretic language, these varieties are Hessenberg varieties for the group GL_n, depending not only on the graph, but on a regular semisimple element in the Lie algebra $\mathfrak{gl}_n$. The formula was proved by Brosnan–Chow [BC18]. In particular, Brosnan–Chow showed how to construct the S_n-action from the monodromy of a family of Hessenberg varieties over the regular semisimple locus of $\mathfrak{gl}_n$.

Hessenberg varieties can be generalized to any reductive algebraic group G, by replacing unit interval graphs with so-called Hessenberg subspaces of the Lie algebra of G. This suggests a generalization of the e-positivity conjecture in terms of Hessenberg varieties, replacing S_n with the Weyl group of G. More daringly, it suggests a generalization of Conjecture A in terms of varieties associated with elements of the Weyl group.

In [AN25], Abreu–Nigro showed that the left-hand side of (1.1) records a monodromy representation on the *intersection* cohomology of a variety $\bar{Y}_{z,g}$ built from z and an arbitrary choice of regular semisimple element $g \in \mathrm{GL}_n(\mathbf{C})$. This variety is now called a Lusztig variety, as it is implicitly defined for general G in Lusztig's papers about character sheaves [Lus85b]. Explicitly, if $\mathcal{B}$ denotes the flag variety of G, and W denotes its Weyl group, then for any $z \in W$ and $g \in G(\mathbf{C})$, the *closed Lusztig variety* $\bar{Y}_{z,g}$ is the Zariski closure in $\mathcal{B}$ of the *open Lusztig variety*

$$Y_{z,g} := \{B \in \mathcal{B} \mid B \xrightarrow{z} g^{-1}Bg\},$$

where $B \xrightarrow{z} B'$ indicates that a pair of Borels (B, B') is in relative position z. Intersection cohomology, rather than singular cohomology, is needed because in general, $\bar{Y}_{z,g}$ may be singular. Abreu–Nigro's result harmonizes with recent work of Brosnan–Hong–Lee [BHL25], showing that when a (smooth) permutation $z \in W$ corresponds to a Hessenberg subspace $\mathfrak{H}$ of the Lie algebra of G, the W-representation arising from a Lusztig variety for z matches that arising from a Hessenberg variety for $\mathfrak{H}$.

For a general reductive group G with Weyl group W, we are led to seek a generalization of Conjecture A to W that replaces the left-hand side of (1.1) with the intersection cohomology of a closed Lusztig variety for G.

[2] Another statement known as the Shareshian–Wachs conjecture.

It turns out that the graded representation of W on the intersection cohomology of $\bar{Y}_{z,g}$ does not depend on g, at least when g is sufficiently generic. In fact, Lusztig's work allows us to give a purely algebraic formula for this representation in terms of a pairing $\{-,-\}\colon \mathrm{Irr}(W) \times \mathrm{Irr}(W) \to \mathbf{Q}$ called the *(truncated) exotic Fourier transform*, originating from his work on characters of finite reductive groups [Lus84b]. We review its definition in Section 2. For W of classical type, Lusztig gives a purely combinatorial algorithm to calculate $\{-,-\}$.

Example 1.1. In the type-A case where $W = S_n$, the exotic Fourier transform is simply the pairing where $\{\chi,\chi\} = 1$ and $\{\chi,\psi\} = 0$ for $\chi \neq \psi$. So $\tau_{z,\mathsf{v}}$ does generalize the left-hand side of (1.1).

Let IH^* denote intersection cohomology with complex coefficients. If g is regular semisimple, then $\mathrm{IH}^i(\bar{Y}_{z,g})$ vanishes for odd degrees i, and its top degree is $\ell(z)$, the Bruhat length of z. Let $G^{rs} \subseteq G$ be the regular semisimple locus, a Zariski dense open subset. For any $z \in W$, let $\tau_{z,\mathsf{v}}\colon W \to \mathbf{Z}[\mathsf{v}^{\pm 1}]$ be defined by

$$\tau_{z,\mathsf{v}} = \sum_{\chi,\psi \in \mathrm{Irr}(W)} \{\chi,\psi\}\psi_{\mathsf{v}}(c_z)\chi.$$

In Section 3, we use Lusztig's results in [Lus85b, Lus86] and the "generic base change" in [Del77] to show:

Theorem 1.2. *Fix a connected complex reductive group G with Weyl group W. Then for any $z \in W$, there is a conjugation-stable, (Zariski) dense open subset $\mathcal{M}_z^{rs} \subseteq G^{rs}$ such that*

$$\mathsf{v}^{\ell(z)}\tau_{z,\mathsf{v}}(w) = \sum_i \mathsf{v}^{2i}\,\mathrm{tr}(w \mid \mathrm{IH}^{2i}(\bar{Y}_{z,g})) \quad \textit{for all } g \in \mathcal{M}_z^{rs}(\mathbf{C}) \textit{ and } w \in W,$$

where the W-action on the intersection cohomology arises from the monodromy action of the fundamental group of G^{rs} based at g.

By Example 1.1, the type-A case of Theorem 1.2 recovers the generic case of Theorem 1.5 in [AN25]. A better way to say this is that Abreu–Nigro's theorem implies that the following conjecture holds when G is semisimple of type A.

Conjecture B. *In Theorem 1.2, we can always take $\mathcal{M}_z^{rs} = G^{rs}$.*

1.3. **The Springer Correspondence.** Next, we seek to generalize the characters ind_μ on the right-hand side of (1.1). It turns out that simply replacing them with the characters of W induced from trivial characters of parabolic subgroups does not work, even in type BC_2, because these are too few. Instead, we turn to an algebro-geometric interpretation of ind_μ that allows more possibilities.

Prior to Lusztig's work on character sheaves, Springer had shown that for any G and unipotent $u \in G(\mathbf{C})$, the Weyl group W acts on the cohomology of

$$\mathcal{B}_u := \{B \in \mathcal{B} \mid u \in B\},$$

even though this action usually does not come from automorphisms of $\mathcal{B}_u$ itself. The variety $\mathcal{B}_u$ is now called a *Springer fiber*. Note that if we write e for the identity of W, then $\mathcal{B}_u = Y_{e,u}$.

Let $C_u \subseteq G$ be the conjugacy class of u and $G_u \subseteq G$ its centralizer. Via the isomorphism of varieties $C_u \simeq G/G_u$, any representation of the component group $A_u := \pi_0(G_u)$ defines a local system on C_u. The cohomology $\mathrm{H}^*(\mathcal{B}_u)$ (again with complex coefficients) can be decomposed into pieces indexed by such local systems. Indeed, the G-action on $\mathcal{B}$ restricts to a G_u-action on $\mathcal{B}_u$, which induces an A_u-action on $\mathrm{H}^*(\mathcal{B}_u)$. The actions of W and A_u centralize each other. For any $\kappa \in \mathrm{Irr}(A_u)$, let $\mathrm{H}^*(\mathcal{B}_u)_\kappa$ denote the multiplicity space of κ in $\mathrm{H}^*(\mathcal{B}_u)$.

For any u, let d_u be the dimension of $\mathcal{B}_u$. The *Springer correspondence* states that each irreducible character of W is afforded by the representation of W on $\mathrm{H}^{2d_u}(\mathcal{B}_u)_\kappa$ for some pair (u, κ), and that this pair is uniquely determined up to conjugacy. It can happen that $\mathrm{H}^{2d_u}(\mathcal{B}_u)_\kappa$ is zero for some (u, κ); in this case, we say that (u, κ), or the associated local system, is *cuspidal*. Lusztig showed [Lus84a] that there are very few cuspidal local systems: If G is almost-simple and adjoint, then there is at most one, and only in types B, C, D, G_2, F_4, E_8.

For any $\psi \in \mathrm{Irr}(W)$, let $spr_\psi \colon W \to \mathbf{Z}$ be defined by

$$spr_\psi(w) = \sum_i \mathrm{tr}(w \mid \mathrm{H}^{2i}(\mathcal{B}_u)_\kappa), \quad \text{where } \psi \mapsto [u, \kappa] \text{ under Springer.} \tag{1.2}$$

(A purity result shows that $\mathrm{H}^i(\mathcal{B}_u)$, like $\mathrm{IH}^*(\bar{Y}_{z,g})$, vanishes for odd i.) It turns out that these functions only depend on W, not G. For $W = S_n$, the groups A_u are all trivial, and spr_{χ_μ} recovers ind_μ for all $\mu \vdash n$ [Lus04]. For other classical types, D. Kim gives a purely combinatorial formula for a coarser function, the character of the total cohomology $\sum_i \mathrm{H}^{2i}(\mathcal{B}_u)$, in terms of modified Kostka polynomials [Kim18]. For general W, the linear map on class functions that sends $\psi \mapsto spr_\psi$ is unipotent, hence invertible.

For any $z \in W$, let $\alpha_z^\psi(\mathsf{v}) \in \mathbf{Z}[\mathsf{v}^{\pm 1}]$ be the (necessarily palindromic) Laurent polynomials indexed by $\psi \in \mathrm{Irr}(W)$ such that

$$\tau_{z,\mathsf{v}} = \sum_{\psi \in \mathrm{Irr}(W)} \alpha_z^\psi(\mathsf{v}) spr_\psi \tag{1.3}$$

as functions on W. Theorem 1.2 shows that these virtual multiplicities determine and are determined by the monodromy of the closed Lusztig varieties $\bar{Y}_{z,g}$ for sufficiently generic $g \in G^{rs}(\mathbf{C})$. In Section 4, we give an alternate characterization of the $\alpha_z^\psi(\mathsf{v})$ that solely involves the unipotent locus $G^{uni} \subseteq G$, at the expense of a detour from the complex numbers to finite fields: See Corollary 4.6. Patrick Brosnan has pointed out that this is reminiscent of how the regular semisimple and unipotent loci of the Lie algebra are related under the Fourier–Deligne transform. A notable byproduct is a formula for the Kazhdan–Lusztig polynomial of (e, z) in terms of the $\alpha_z^\psi(\mathsf{v})$: See Corollary 4.9.

It would be pleasant to generalize Conjecture A by generalizing Haiman's $\alpha_z^\mu(\mathsf{v})$ to $\alpha_z^\psi(\mathsf{v})$. Unfortunately, the nonzero coefficients of $\alpha_z^\psi(\mathsf{v})$ can fail to be positive. In fact, they can have mixed signs. They can also fail to form a unimodal sequence, even in absolute value.

We do see hints of certain patterns. We will see in Section 5 that for W of low rank, the positive unimodal cases largely predominate. The failures occur only when ψ belongs to a fixed, relatively small subset $\mathrm{Irr}(W)' \subseteq \mathrm{Irr}(W)$. However, $\mathrm{Irr}(W)'$ seems to grow as the rank of W grows.

The elements of $\mathrm{Irr}(W)'$ tend to correspond, under Springer, to local systems on unipotent conjugacy classes of relatively large dimension. For instance, the character of the reflection representation, which maps to the trivial local system on the subregular class, often belongs to $\mathrm{Irr}(W)'$. Else, its Lusztig–Spaltenstein dual [Car93, §13.4] often belongs to $\mathrm{Irr}(W)'$.

Question 1.3. Let $\mathrm{Irr}(W)'$, *resp.* $\mathrm{Irr}(W)''$, be the subset of characters $\psi \in \mathrm{Irr}(W)$ such that for some $z \in W$, the nonzero coefficients of $\alpha_z^\psi(\mathsf{v})$ fail to be positive, *resp.* unimodal in absolute value.

- Do we always have $\mathrm{Irr}(W)'' \subseteq \mathrm{Irr}(W)'$?
- Is the ratio $|\mathrm{Irr}(W)'|/|\mathrm{Irr}(W)|$, *resp.* $|\mathrm{Irr}(W)''|/|\mathrm{Irr}(W)|$, uniformly bounded away from 1?
- Is there a uniform characterization of $\mathrm{Irr}(W)'$, *resp.* $\mathrm{Irr}(W)''$?

1.4. **Lusztig's Families.** Another structure from Lusztig's work, related both to the exotic Fourier transform defining $\tau_{z,\mathsf{v}}$ and to the Springer correspondence, is his notion of *families*, also reviewed in Section 2. We will only discuss their truncations to $\mathrm{Irr}(W)$, which may be viewed as the sets $\mathcal{F}$ providing the finest partition $\mathrm{Irr}(W) = \coprod_{\mathcal{F}} \mathcal{F}$ with respect to which $\{-,-\}$ is block-diagonal. These truncated families may also be defined combinatorially, in terms of the two-sided cells of W.

Each family of $\mathrm{Irr}(W)$ contains a unique *special character* [Car93, §11.3]. Under the Springer correspondence, these are always sent to trivial local systems. Using these facts, Lusztig showed that the partition of W into subsets $\mathcal{F}$ gives rise to a remarkable partition of G^{uni} into associated subsets $G^{uni}_{\mathcal{F}}$, now called *special pieces* [Lus97]. More precisely, the family $\mathcal{F}$ is precisely the set of irreducible characters of W mapped to local systems on conjugacy classes $C \subseteq G^{uni}_{\mathcal{F}}$.

We are led to study the role of Lusztig's families in (1.3). For any $z \in W$ and (truncated) family $\mathcal{F} \subseteq \mathrm{Irr}(W)$, form the unnormalized average

$$\alpha_z^{\mathcal{F}}(\mathsf{v}) = \sum_{\psi \in \mathcal{F}} \alpha_z^\psi(\mathsf{v}). \tag{1.4}$$

By Example 1.1, the families for $W = S_n$, are precisely the singleton subsets of $\mathrm{Irr}(W)$, so the Laurent polynomials $\alpha_z^{\mathcal{F}}(\mathsf{v})$ reduce to the $\alpha_z^\psi(\mathsf{v})$, which in turn reduce to Haiman's $\alpha_z^\mu(\mathsf{v})$.

In low rank, many of the negative signs that we observe in the $\alpha_z^{\psi}(\mathsf{v})$ are washed away miraculously when we collect them into the $\alpha_z^{\mathcal{F}}(\mathsf{v})$, enough for us to propose the following generalization of Conjecture A to type BC.

Conjecture C. *Suppose that W is a Weyl group of type BC_n for some n. Then for any $z \in W$ and family $\mathcal{F} \subseteq \mathrm{Irr}(W)$, the nonzero coefficients of $\alpha_z^{\mathcal{F}}(\mathsf{v})$ are all positive and form a unimodal sequence.*

Yet in other types, starting with F_4, the nonzero coefficients of $\alpha_z^{\mathcal{F}}(\mathsf{v})$ still fail to be positive or unimodal or both in a handful of cases.

1.5. **Sortable Elements.** Searching for something more robust, we re-examine the notion of 312-avoiding permutation that links Conjecture A with the conjectures of Shareshian–Wachs and Stanley–Stembridge.

The property of being 312-avoiding, for the elements of S_n, is a special case of a property that can be defined for the elements of any irreducible Coxeter group W with a chosen standard Coxeter element $c \in W$: that of being *c-sortable*. To explain, fix a reduced word $\vec{c} = (s^{(1)}, \ldots, s^{(h)})$ for c: a certain ordering of the set of simple reflections $S \subseteq W$. Then any $w \in W$ defines a sequence of subsets of S, according to how the elements of the lexicographically first subword of

$$\vec{c}^{\infty} := s^{(1)}, \ldots, s^{(h)}, \mid s^{(1)}, \ldots, s^{(h)} \mid s^{(1)}, \ldots, s^{(h)}, \mid \ldots$$

sit between the dividers $|$. Following Reading [Rea07], w is called *c-sortable* when these subsets of S are decreasing under inclusion, as we scan to the right: This depends only on c, not $\vec{c}$. To be 312-avoiding in S_n is to be c-sortable for the n-cycle $c = s_1 s_2 \cdots s_{n-1}$, where s_i swaps i and $i+1$, and permutations are composed from right to left.

When the rank of W is low, the elements that are c-sortable for some c comprise most of the group, but as the rank grows, they rapidly thin out. In particular, $W(F_4)$ has only 532 such elements out of 1152 total. We can finally state a uniform conjecture.

Conjecture D. *Suppose that W is an irreducible finite Weyl group. Then there is at least one standard Coxeter element $c \in W$ such that, for any c-sortable $z \in W$ and family $\mathcal{F} \subseteq \mathrm{Irr}(W)$, the nonzero coefficients of $\alpha_z^{\mathcal{F}}(\mathsf{v})$ are all positive and form a unimodal sequence.*

Just as Conjecture C generalizes Conjecture A to type BC, so Conjecture D generalizes the 312-avoiding case of Conjecture A, the Shareshian–Wachs conjecture, to all Weyl groups.

In Section 5, we summarize the verification of Conjectures C–D in low rank, relying on the CHEVIE package for GAP3 [GHL+96].

Theorem 1.4. *Conjectures C–D hold whenever the rank of W is at most 5.*

The examples will show that when z is merely c-sortable for *some* c, the nonzero coefficients of $\alpha_z^{\psi}(\mathsf{v})$ or $\alpha_z^{\mathcal{F}}(\mathsf{v})$ can fail to be positive, or unimodal in absolute value, for some ψ or $\mathcal{F}$. It can also happen that for every c, there is some choice of c-sortable z and ψ such that positivity fails for $\alpha_z^{\psi}(\mathsf{v})$. In this sense, Conjecture D appears to be the strongest generalization we can make.

1.6. **Further Directions.** There are at least two.

1.6.1. *From Finite to Affine.* Affinization is the principle that structures for a complex reductive group G often have analogues for its loop group LG or the associated affine Kac–Moody group.

The affine analogue of a Lusztig variety was first studied in [Lus11b] and [KV12]. The term *affine Lusztig variety* was introduced by X. He [He25], although other names have been used, such as *generalized affine Springer fiber*, *multiplicative affine Springer fiber*, and *Kottwitz–Viehmann variety*. Just as Lusztig varieties originally appear in the theory of character sheaves, so affine Lusztig varieties are expected to appear in a putative theory of character sheaves on loop groups.

Since affine Lusztig varieties for regular semisimple elements of the loop group LG are still finite-dimensional, we can use them to define an analogue of $\tau_{z,\mathsf{v}}$ geometrically. It should take the form of a class function on an extended affine Weyl group. However, it is less clear what should replace the functions spr_{ψ}, since affine Springer fibers for unipotent elements of LG are infinite-dimensional. It seems plausible that one should use the ungraded characters of affine Springer representations arising from *topologically* unipotent elements of LG. Here, the analogues of the groups A_u and their characters κ play a major role in the arithmetic of the affine Springer fibers: more precisely, in the theory of endoscopy in the Langlands program [Wan26].

Question 1.5. Is there an analogue of Conjecture D, relating the intersection cohomology of regular semisimple affine Lusztig varieties to (ungraded) affine Springer characters?

1.6.2. *From Weyl Groups to Coxeter Groups.* In [Lus93, Lus94], Lusztig and Malle showed that certain postulates uniquely determine an analogue of the exotic Fourier transform from [Lus84b] for any finite Coxeter group. (More accurately, Malle's appendix to [Lus94] handles type H_4.) Consequently, $\tau_{z,\mathsf{v}}$ can be generalized from Weyl groups to all finite groups.

At the same time, in [AA08], Achar–Aubert showed that certain axioms determine what they call a *preferred* analogue of the map $\psi \mapsto spr_{\psi}$ for dihedral groups. In [Ach09], building on prior observations in [GM99], Achar reports on attempts to extend the characters spr_{ψ} even further, to *complex* reflection groups.

Finally, Lusztig observed in [Lus15, §10] that there is a uniform way to generalize his families (and their truncations to $\mathrm{Irr}(W)$ to finite Coxeter groups, using categories of Soergel bimodules.

Question 1.6. Is there a uniform generalization of Conjecture D to all finite Coxeter groups?

1.7. **Acknowledgments.** I thank Claude (Sonnet 4.6) for writing Python scripts to help me analyze CHEVIE data. I also thank Patrick Brosnan and Rachel McEnroe for much encouragement.

2. Deligne–Lusztig Theory

In this brief section, we review various ideas from [DL76] and Lusztig's subsequent work on the characters of finite reductive groups [Lus84b], including definitions of the pairing $\{-,-\}$ mentioned in §1.2 and the notion of families mentioned in §1.5.

Let $\mathbf{k}$ be the algebraic closure of a finite field $\mathbf{F}$. Let G be a connected reductive algebraic group over $\mathbf{k}$ with Weyl group W, such that the characteristic of $\mathbf{k}$ is a good prime for G [Car93, 28]. Let $F\colon G \to G$ be the Frobenius map corresponding to the split $\mathbf{F}$-structure on G.

For all $w \in W$, the Deligne–Lusztig variety X_w is defined by the pullback square:

$$\begin{tikzcd} O_w \arrow[d] & X_w \arrow[l] \arrow[d] \\ \mathcal{B} \times \mathcal{B} & \mathcal{B} \arrow[l, "\mathrm{Id}\times F"'] \end{tikzcd}$$

In other words, $X_w \subseteq \mathcal{B}$ is the subvariety of Borels B such that $B \xrightarrow{w} F(B)$. The finite reductive group G^F acts on X_w, and hence on its compactly-supported ℓ-adic étale cohomology $\mathrm{H}^i_c(X_w, \bar{\mathbf{Q}}_\ell)$. The *Deligne–Lusztig character* $R_w\colon G^F \to \bar{\mathbf{Q}}_\ell$ is the virtual character defined by

$$R_w(g) = \sum_i (-1)^i \operatorname{tr}(g \mid \mathrm{H}^i_c(X_w, \bar{\mathbf{Q}}_\ell)).$$

For any $\psi \in \mathrm{Irr}(W)$, the *almost-character* $R_\psi\colon G^F \to \bar{\mathbf{Q}}_\ell$ is defined by

$$R_\psi = \frac{1}{|W|} \sum_{w \in W} \psi(w) R_w.$$

As in Section 1, let e be the identity of W. Then X_e is the finite set $\mathcal{B}^F$, and R_e is the character of the principal series representation of G^F on $\mathrm{H}^0(\mathcal{B}^F, \bar{\mathbf{Q}}_\ell)$.

Let $q = |\mathbf{F}|$. Fix a square root $q^{1/2} \in \bar{\mathbf{Q}}_\ell^\times$. By a classical theorem of Iwahori, the specialized Hecke algebra $H_W(q) := H_W(\mathsf{v})|_{\mathsf{v} \to q^{1/2}}$ acts on $\mathrm{H}^0(\mathcal{B}^F, \bar{\mathbf{Q}}_\ell)$ by operators commuting with G^F. Namely, $q^{1/2}\sigma_w$ sends the indicator function at a Borel B to the indicator on the set of Borels B' such that $B \xrightarrow{w} B'$. Therefore, R_e can be promoted to a bitrace $R_e\colon G^F \times H_W(q) \to \bar{\mathbf{Q}}_\ell$. We have a decomposition

$$R_e = \sum_{\chi \in \mathrm{Irr}(W)} \rho_\chi \otimes \chi_{\mathsf{v}}|_{\mathsf{v} \to q^{1/2}}, \tag{2.1}$$

where ρ_χ is an irreducible character of G^F called the *unipotent principal series character* associated with χ.

Let $(-,-)_{G^F}$ be the scalar product on class functions on G^F. We define the *(truncated) exotic Fourier transform* $\{-,-\}$ by setting

$$\{\chi,\psi\} = (\rho_\chi, R_\psi)_{G^F} \quad \text{for all } \chi,\psi.$$

It turns out that this, indeed, only depends on W, and not on $\mathbf{F}$ or G.

An irreducible character of G^F is called *unipotent* when it has nonzero scalar product with R_w for some $w \in W$. Let $\mathrm{Uch}(G^F) \subseteq \mathrm{Irr}(G^F)$ be the set of unipotent irreducible characters. Let $\approx$ be the transitive closure of the reflexive, symmetric relation on $\mathrm{Uch}(G^F)$ where $\rho \approx \rho'$ if and only if ρ and ρ' both have nonzero scalar product with R_ψ for some $\psi \in \mathrm{Irr}(W)$. Lusztig's *families* of unipotent irreducible characters are the equivalence classes of $\mathrm{Uch}(G^F)$ under $\approx$ [Car93, §12.3].

The map $\chi \mapsto \rho_\chi$ gives us an embedding of $\mathrm{Irr}(W)$ into $\mathrm{Uch}(G^F)$. The *(truncated) families* of $\mathrm{Irr}(W)$ are the restrictions of the families above to $\mathrm{Irr}(W)$. For an alternative definition based on the two-sided cells of W, see [Car93, §12.4].

3. The Regular Semisimple Locus

The goal of this section is to prove Theorem 1.2.

Note that, while Section 1 discussed algebraic varieties over the complex numbers, Lusztig's papers on character sheaves start from a general algebraically closed field $\mathbf{k}$. We shall do the same, and specialize $\mathbf{k}$ to the field of complex numbers or to the algebraic closure of a finite field only when needed. We fix a prime ℓ invertible in $\mathbf{k}$, and work with $\bar{\mathbf{Q}}_\ell$-complexes in the étale topology on $\mathbf{k}$-varieties.

3.1. **Setup.** As in Sections 1–2, G is a connected reductive algebraic group over $\mathbf{k}$ with Weyl group W. The characteristic of $\mathbf{k}$ is either zero or a good prime for G.

Let $\mathcal{B}$ be the flag variety of G, and let $act\colon \mathcal{B}\times G \to \mathcal{B}\times\mathcal{B}$ be the map $act(B,g) = (B, g^{-1}Bg)$. For all $v \in W$, let $O_z \subseteq \mathcal{B}\times\mathcal{B}$ be the G-orbit of pairs in relative position w. Sheaves on these orbits form a geometrization of the Hecke algebra H_W, while pullback-pushforward through the correspondence

$$\mathcal{B}\times\mathcal{B} \xleftarrow{act} \mathcal{B}\times G \xrightarrow{pr} G$$

is a geometrization of its cocenter map.

Let $Y_z \subseteq \mathcal{B}\times G$ be defined by the pullback square:

$$\begin{tikzcd} O_z \arrow[d] & Y_z \arrow[l] \arrow[d] \\ \mathcal{B}\times\mathcal{B} & \mathcal{B}\times G \arrow[l, "act"'] \end{tikzcd}$$

Let $\bar{Y}_z$ be the Zariski closure of Y_z in $\mathcal{B}\times G$. Thus $\bar{Y}_z = \bigcup_{w\le v} Y_w$, where $\le$ is the Bruhat order on W corresponding to the closure order on the orbits O_w.

Let $j\colon Y_z \to \bar{Y}_z$ be the inclusion map, and let $j_{!*}$, denote intermediate extension of shifted perverse sheaves along j [Ach21, §5.1]. Let

$$J_z = j_{!*}\underline{\bar{\mathbf{Q}}_\ell}_{Y_z},$$

the (unshifted) intersection cohomology complex of $\bar{Y}_z$. Let

$$\bar{\pi}_z \colon \bar{Y}_z \to G$$

be the composition $\bar{Y}_z \to \mathcal{B} \times G \xrightarrow{pr} G$, a proper map, and let

$$\bar{K}_z = \bar{\pi}_{z,!} J_z = \bar{\pi}_{z,*} J_z.$$

The notations $\bar{Y}_z, J_z, \bar{\pi}_z, \bar{K}_z$ are based on Lusztig's notations in [Lus85b]. By the decomposition theorem [Ach21, §3.9], there is a noncanonical isomorphism

$$\bar{K}_z \simeq \bigoplus_i {}^p\mathcal{H}^i(\bar{K}_z)[-i],$$

where ${}^p\mathcal{H}^i$ denotes the ith cohomology sheaf with respect to the perverse t-structure, and each perverse sheaf ${}^p\mathcal{H}^i(\bar{K}_z)$ is semisimple. The simple summands of these perverse sheaves, as we run over z and i, are called *unipotent character sheaves*.

3.2. **The Grothendieck–Springer Sheaf.** The map $\bar{\pi}_e \colon Y_e = \bar{Y}_e \to G$ is the *Grothendieck–Springer alteration* of G. Below, we review facts about $\bar{\pi}_e$ that can be proved in an analogous way to their counterparts for the Lie algebra of G [Ach21, §8.2].

First, the pullback of $\bar{\pi}_e$ to the regular semisimple locus G^{rs} is a Galois étale cover with Galois group W. In particular, if $\mathbf{k} = \mathbf{C}$, then W is a quotient of the topological fundamental group of G^{rs}.

Next, $\bar{\pi}_e$ is a small map. So $\bar{K}_e$ is the intermediate extension of its restriction $\bar{K}_e|_{G^{rs}}$, and the Galois action of W on $\bar{K}_e|_{G^{rs}}$ induces one on $\bar{K}_e$ by functoriality. We obtain a W-equivariant decomposition

$$\bar{K}_e[\dim G] \simeq \bigoplus_{\chi \in \mathrm{Irr}(W)} V_\chi \otimes \mathcal{A}_\chi, \tag{3.1}$$

where the $\mathcal{A}_\chi$ are simple perverse sheaves and $V_\chi := \mathrm{Hom}(\mathcal{A}_\chi, \bar{K}_e[\dim G])$ affords a representation of W with character χ.

Lemma 3.1 (Lusztig)**.** *Suppose that* $\mathbf{k}$ *is either a subfield of* $\mathbf{C}$ *or the algebraic closure of a finite field. Then for any* $z \in W$ *and* $\chi \in \mathrm{Irr}(W)$*, we have*

$$\sum_i (-\mathsf{v})^i (\mathcal{A}_\chi : {}^p\mathcal{H}^i(\bar{K}_z)) = (-1)^{\dim G} \mathsf{v}^{\dim G + \ell(z)} \sum_{\psi \in \mathrm{Irr}(W)} \{\chi, \psi\} \psi_{\mathsf{v}}(c_z),$$

where ℓ *is Bruhat length, as in §1.2, and* $\{-,-\}$ *is the exotic Fourier transform, as in Section 2.*

Proof. When G is defined over a finite field and $\mathbf{k}$ is its algebraic closure, this follows from Lusztig's [Lus85b, Corollary 14.11] and [Lus86, Theorem 23.1(c)]. See also the explanation of [WW11, 413].

When $\mathbf{k} \subseteq \mathbf{C}$, we deduce the result by a standard "spreading out" argument, following [Spr78, §2.7] and the references to SGA 4 there. In brief, we spread out

G and the relevant $\bar{\mathbf{Q}}_\ell$-complexes to a model over a finitely generated ring $R \subseteq \mathbf{C}$, using the specialization theorem for étale cohomology. After shrinking R, we may assume that the complexes are locally constant over $\mathrm{Spec}(R)$. At some closed point $\mathfrak{p} \in \mathrm{Spec}(R)$ and geometric point $\bar{\mathfrak{p}}$ over $\mathfrak{p}$, the multiplicity formula holds at $\bar{\mathfrak{p}}$. Hence, by the comparison theorem for étale cohomology, the same formula holds at the geometric point corresponding to the embedding of R into $\mathbf{C}$. □

For any constructible complex K on G, we write $\mathcal{H}^*_g(K)$ to denote the stalk at g of its cohomology $\mathcal{H}^*(K)$.

Lemma 3.2. *Suppose that $\mathbf{k} = \mathbf{C}$. Then for any $\chi \in \mathrm{Irr}(W)$ and $g \in G^{rs}(\mathbf{C})$, the graded vector space $\mathcal{H}^*_g(\mathcal{A}_\chi)$ is concentrated in degree $-\dim G$. The monodromy action of $\pi_1(G^{rs}, g)$ on it defines a representation of W with character χ.*

Proof. Let $Y_e^{rs} \to G^{rs}$ be the pullback of $Y_e \to G$. Since $Y_e^{rs} \to G^{rs}$ is a W-torsor, and W is finite, we know that $\mathcal{H}^*_g(\bar{K}_e)$ is concentrated in degree 0. Since $\mathcal{A}_\chi$ is a summand of $\bar{K}_e[\dim G]$, we deduce the first claim.

By construction, the monodromy action on the W-torsor $\mathcal{H}^0_g(\bar{K}_e)$ factors through the regular representation of W. The resulting W-action commutes with the one arising from the Galois action on $\bar{K}_e$. So, from comparing (3.1) to the bi-equivariant decomposition $\bar{\mathbf{Q}}_\ell W \simeq \bigoplus_\chi V_\chi \otimes V_{\chi^\vee}$, we deduce that the monodromy action on $\mathcal{H}^*_g(\mathcal{A}_\chi)$ has character $\chi^\vee$, where $(-)^\vee$ denotes duals. But irreducible characters of Weyl groups are real-valued, so they are self-dual. □

3.3. **Cuspidal Terms.** The perverse sheaves ${}^p\mathcal{H}^i(\bar{K}_z)$ can have summands not isomorphic to $\mathcal{A}_\chi$ for any χ. These unipotent character sheaves are said to be *cuspidal*, parallel to the terminology from the introduction.

Lemma 3.3 (Lusztig)**.** *If A is a cuspidal unipotent character sheaf on G, and $g \in G(\mathbf{k})$ is regular semisimple, then $\mathcal{H}^*_g(A)$ is zero.*

Proof. Let $Z_G \subseteq G$ be the center and $Z_G^\circ \subseteq Z_G$ its neutral component.

By [Lus85a, (7.1.2)], $A = \mathrm{IC}(\bar{\Sigma}, \mathcal{E})[d]$ for some $(G \times Z_G^\circ)$-orbit $\Sigma \subseteq G$, some simple $(G \times Z_G^\circ)$-equivariant local system $\mathcal{E}$ on Σ, and some integer d. Moreover, by [Lus86, Theorem 23.1(a)] (using definitions from [Lus85b, (13.9.2)] and [Lus85a, (7.7)]), A is *clean*: Its restriction to the complement $\bar{\Sigma} - \Sigma$ is zero. So it remains to show that Σ cannot contain g.

If G is a torus, then the only unipotent character sheaf is the constant sheaf, so G cannot be a torus. So G_g/Z_G° must be nontrivial, where $G_g \subseteq G$ is the centralizer of g. But if $g \in \Sigma$, then [Lus85a, (7.1.2)] shows that G_g/Z_G° must be unipotent, whereas G_g is a torus. □

3.4. **The Monodromy Action.** For the rest of this section, $\mathbf{k} = \mathbf{C}$. Recall that $\bar{Y}_{z,g}$ is the fiber of $\bar{Y}_z$ above $g \in G(\mathbf{C})$, and that $\mathrm{IH}^*(Y_{z,g})$ is its intersection cohomology with coefficients in $\mathbf{C}$. We fix an isomorphism $\mathbf{C} \simeq \bar{\mathbf{Q}}_\ell$, so that we can identify the intersection cohomology from Section 1, which used $\mathbf{C}$-coefficients, with the (hyper)cohomology of a $\bar{\mathbf{Q}}_\ell$-complex.

Lemma 3.4. *For any $z \in W$, there is a conjugation-stable, (Zariski) dense open subset $\mathcal{M}_z \subseteq G$ such that*

$$\mathrm{IH}^*(\bar{Y}_{z,g}) \simeq \mathcal{H}^*_g(\bar{K}_z) \quad \textit{for all } g \in \mathcal{M}_z(\mathbf{C}).$$

Proof. Let $i_g \colon \bar{Y}_{z,g} \to \bar{Y}_z$ be the inclusion map, and let i_g^* be ordinary (derived) pullback along i_g. Let $j_g \colon Y_{z,g} \to \bar{Y}_{z,g}$ be the inclusion map, parallel to our earlier notation $j \colon Y_z \to \bar{Y}_z$, and let $j_{g,!*}$, denote intermediate extension along j_g. Then

$$\begin{aligned} \mathrm{IH}^*(\bar{Y}_{z,g}) &= \mathrm{H}^*(\bar{Y}_{z,g}, j_{g,!*} i_g^* \underline{\mathbf{C}}_{Y_z}), \\ \mathcal{H}^*_g(\bar{K}_z) &= \mathrm{H}^*(\bar{Y}_{z,g}, i_g^* j_{!*} \underline{\mathbf{C}}_{Y_z}). \end{aligned}$$

So it suffices to show that $j_{g,!*} i_g^* \simeq i_g^* j_{!*}$ as functors from the constructible derived category of Y_z to that of $\bar{Y}_{z,g}$. This holds on some dense open subset of G by the "generic base change" theorem of [Del77, Théorème 1.9].[3]

Finally, if g and g' are conjugate, then conjugation provides an isomorphism from the diagram $Y_{z,g} \to \bar{Y}_{z,g} \to \bar{Y}_z$ to the diagram $Y_{z,g'} \to \bar{Y}_{z,g'} \to \bar{Y}_z$. So the above isomorphism of functors is stable under replacing g with g'. □

Let $\mathcal{M}_z^{rs} = G^{rs} \cap \mathcal{M}_z$, for the locus $\mathcal{M}_z$ in Lemma 3.4. (Recall that G^{rs} is also a conjugation-stable, dense open subset of G.)

Proof of Theorem 1.2. Fix $g \in \mathcal{M}_z^{rs}(\mathbf{C})$ and $w \in W$. We are claiming that

$$\mathsf{v}^{\ell(z)} \tau_{z,\mathsf{v}}(w) = \sum_i \mathsf{v}^{2i} \operatorname{tr}(w \mid \mathrm{IH}^{2i}(\bar{Y}_{z,g})), \tag{3.2}$$

where the W-action on $\mathrm{IH}^*(\bar{Y}_{z,g})$ arises from the monodromy action of $\pi_1(G^{rs}, g)$. In the ring of formal power series in v over the (complexified) representation ring of W, we have

$$\begin{aligned} &\sum_j (-\mathsf{v})^j \, \mathrm{IH}^j(\bar{Y}_{z,g}) \\ &= \sum_j (-\mathsf{v})^j \, \mathcal{H}^j_g(\bar{K}_z) && \text{by Lemma 3.4} \\ &= \sum_{i,j} (-\mathsf{v})^j \, \mathcal{H}^{j-i}_g({}^p\mathcal{H}^i(\bar{K}_z)) && \text{by the decomp. thm.} \\ &= \sum_k (-\mathsf{v})^k \, \mathcal{H}^k_g \left(\sum_i (-\mathsf{v})^i \, {}^p\mathcal{H}^i(\bar{K}_z) \right) \\ &= \sum_k (-\mathsf{v})^k \, \mathcal{H}^k_g \left(\sum_{i,\chi} (-\mathsf{v})^i (\mathcal{A}_\chi : {}^p\mathcal{H}^i(\bar{K}_z)) \mathcal{A}_\chi \right) && \text{by Lemma 3.3} \\ &= \mathsf{v}^{\ell(z)} \sum_{k,\chi,\psi} (-\mathsf{v})^{k+\dim G} \{\chi, \psi\} \psi_{\mathsf{v}}(c_z) \, \mathcal{H}^k_g(\mathcal{A}_\chi) && \text{by Lemma 3.1.} \end{aligned}$$

[3] See also https://mathoverflow.net/a/123787.

By Lemma 3.2, $\mathcal{H}_g^*(\mathcal{A}_\chi)$ is concentrated in degree $-\dim G$, and the monodromy action on it factors through a representation of W with character χ. Combining this with the preceding chain of equalities,

$$\mathsf{v}^{\ell(z)} \sum_{\chi,\psi} \{\chi,\psi\}\psi_{\mathsf{v}}(c_z)\chi(w) = \sum_j (-\mathsf{v})^j \operatorname{tr}(w \mid \mathrm{IH}^j(\bar{Y}_{z,g})).$$

The left-hand side is $\mathsf{v}^{\ell(z)}\tau_{z,\mathsf{v}}(w)$.

Finally, Theorem 1.6(1) and display (9.3) in [AN25] together show that when g is regular semisimple, $\mathrm{IH}^*(\bar{Y}_{z,g})$ vanishes in odd degrees and $\dim \bar{Y}_{z,g} = \dim Y_{z,g} = \ell(z)$, as mentioned in §1.2. So the right-hand side of the last display simplifies to the right-hand side of (3.2). □

Remark 3.5. Fix a parabolic subgroup $W' \subseteq W$. For any character $\psi \in \mathrm{Irr}(W)$, let $\psi^{W'}$ be the scalar product of $\psi|_{W'}$ with the trivial character of W'. Abreu–Nigro's [AN25, Theorem 1.5] asserts that

$$\mathsf{v}^{\ell(z)} \sum_{\psi} \psi_{\mathsf{v}}(c_z)\psi^{W'} = \sum_i \mathsf{v}^{2i} \dim \mathrm{IH}^{2i}(\bar{Y}_{z,g})^{W'} \quad \text{for all } z \in W \text{ and } g \in G^{rs}(\mathbf{C}).$$

(Note that they do not need to restrict to a further locus $\mathcal{M}_z^{rs} \subseteq G^{rs}$.) By comparing with Theorem 1.2, and using the fact that the character table of the Hecke algebra specializes to that of W, which is invertible, we deduce the following curious property of the exotic Fourier transform:

$$\sum_{\chi} \{\chi,\psi\}\chi^{W'} = \psi^{W'} \quad \text{for all } \psi \in \mathrm{Irr}(W). \tag{3.3}$$

Compare to the *tower equivalence* property of $\{-,-\}$ discussed in [Mic23].

4. The Unipotent Locus

The goal of this section is to explain how to reformulate Conjecture D solely in terms of the unipotent locus $G^{uni} \subseteq G$, using ideas from [Tri21].

Throughout this section, we take the setup of Section 2, where $\mathbf{k}$ is the algebraic closure of a finite field $\mathbf{F}$ of size q, and $F\colon G \to G$ is the Frobenius map defining the split $\mathbf{F}$-structure on G. As before, the characteristic of $\mathbf{k}$ is a good prime for G.

4.1. **Springer Fibers and Green Functions.** Below, we review various facts about Springer fibers and functions on G^{uni}. In the literature, most are stated with the nilpotent cone of the Lie algebra of G in place of G^{uni}. However, one can choose a G-equivariant isomorphism between the nilpotent and unipotent loci, matching Springer fibers [Car93, §1.15].

Recall that in the notation of Section 3, the *Springer fiber* above $g \in G$ is

$$\mathcal{B}_g := Y_{e,g} = \bar{Y}_{e,g}.$$

Hence, the W-action on $\bar{K}_e$ induces a W-action on the ℓ-adic étale cohomology $\mathrm{H}^*(\mathcal{B}_g) = \mathcal{H}_g^*(\bar{K}_e)$.

Suppose that $g \in G^F$. Then there is an F-action on $\mathrm{H}^*(\mathcal{B}_g)$ induced by the Weil structure on $\bar{K}_e$ extending that on $\bar{K}_e|_{G^{rs}} = \underline{\bar{\mathbf{Q}}}_{\ell G^{rs}}$. Since F defines the split $\mathbf{F}$-structure on G, the W-action on $\mathrm{H}^*(\mathcal{B}_g)$ commutes with this F-action. Furthermore, writing $G_g \subseteq G$ for the centralizer of g and A_g for its component group, as in §1.3, we see that the G_g-action on $\mathcal{B}_g$ induces an A_g-action on $\mathrm{H}^*(\mathcal{B}_g)$ commuting with both W and F.

Suppose that $u \in (G^{uni})^F$. By [Spr84, Corollary 1], $\mathrm{H}^i(\mathcal{B}_u)$ has no odd étale cohomology for any $u \in (G^{uni})^F$. Moreover, if q is large enough, then by [Spr84, Theorem 2], there is a function $Q_{w,\mathsf{q}} \colon (G^{uni})^F \to \mathbf{Z}[\mathsf{q}]$ such that

$$Q_{w,q}(u) := Q_{w,\mathsf{q}}(u)|_{\mathsf{q}\to q} = \sum_i \mathrm{tr}(wF \mid \mathrm{H}^{2i}(\mathcal{B}_u)).$$

The function $Q_{w,\mathsf{q}}$ is constant along G^F-conjugacy classes. Moreover, it only depends on w up to conjugacy. In the literature, it is sometimes called the *Green function* associated with w.

Recall the simple perverse sheaves $\mathcal{A}_\chi$ discussed in §3.2. For any $\chi \in \mathrm{Irr}(W)$, let $Q_{\chi,q} \colon (G^{uni})^F \to \mathbf{Q}$ be defined by

$$Q_{\chi,q} = \frac{1}{|W|} \sum_{w \in W} \chi(w) Q_{w,q}.$$

Then $Q_{\chi,q}$ is precisely the function recording the trace of Frobenius on the stalks of $\mathcal{H}^*(\mathcal{A}_\chi)[-\dim G]|_{G^{uni}}$. As $\mathcal{A}_\chi|_{G^{uni}}$ remains a simple perverse sheaf, we have

$$\mathcal{A}_\chi|_{G^{uni}} = j_{!*}\mathcal{L}[\dim C] \tag{4.1}$$

for some simple local system $\mathcal{L}$ on some unipotent conjugacy class $j \colon C \to G$. Let $Q^!_{\chi,q}$ be the analogue of $Q_{\chi,q}$ that records the trace of Frobenius on $j_!\mathcal{L}$ instead. Then the collection of functions $\{Q_{\chi,q}\}_\chi$ and $\{Q^!_{\chi,q}\}_\chi$ can be transformed into each other by $\mathbf{Q}$-linear transformations that depend only on W, not on q. The can be determined by an algorithm due to Lusztig and Shoji [Lus86, Sho88].

4.2. **The Springer Correspondence.** One may define the *Springer correspondence* for W as the map from $\mathrm{Irr}(W)$ to the set

$$\left\{ (C, \mathcal{L}) \;\middle|\; \begin{array}{l} C \text{ is an } F\text{-stable unipotent conjugacy class of } G, \\ \mathcal{L} \text{ is a simple local system on } C \text{ up to isomorphism} \end{array} \right\}. \tag{4.2}$$

that sends a character χ to the unique pair $(C, \mathcal{L})$ giving rise to $\mathcal{A}_\chi$ via (4.1). Springer showed that this map is injective [Spr76, Spr78].

To see how this aligns with the description in §1.3: If C is any conjugacy class, and $g \in C(\mathbf{k})$, then isomorphism classes of simple local systems on C correspond to irreducible characters of A_g. In particular, (4.2) is in bijection with the set

$$\Upsilon := \left\{ (u, \kappa) \;\middle|\; \begin{array}{l} u \in G^{uni}(\mathbf{k}), \\ \kappa \in \mathrm{Irr}(A_u) \end{array} \right\} \Big/ G\text{-conjugacy}.$$

For further discussion of how our present setup is related to the one over the complex numbers, see §4.5.

We will use the Springer correspondence to describe the values of the functions $Q^!_{\chi,q}\colon G^F \to \mathbf{Q}$ more explicitly. Like the functions $Q_{\chi,q}$, they are linear combinations of the functions $Q_{w,q}$, so they are constant along G^F-conjugacy classes.

For any $g \in G$, we write C_g to denote its conjugacy class, as in §1.3. If $g \in G^F$, then C is F-stable. But in general, the G^F-conjugacy class of g may be smaller than C_g^F, the set of F-fixed points of C_g. Indeed, Galois cohomology and Lang's theorem yield a bijection from the set of conjugacy classes of A_g onto the set of G^F-conjugacy classes that make up C_g^F.

Suppose that $u \in (G^{uni})^F$. Fix another element $u' \in C_u^F$. If $a(u,u')$ represents the conjugacy class of A_u indexing the G^F-conjugacy class of u', then under any isomorphism $\mathcal{B}_u \simeq \mathcal{B}_{u'}$ induced by conjugation, the F-action on $\mathrm{H}^*(\mathcal{B}_u)$ corresponds to the $a(u,u')F$-action on $\mathrm{H}^*(\mathcal{B}_{u'})$. Treating the irreducible characters of A_u as $\bar{\mathbf{Q}}$-valued functions, and writing $\mathrm{H}^*(\mathcal{B}_u)_\kappa$ for the multiplicity space of $\kappa \in \mathrm{Irr}(A_u)$ in $\mathrm{H}^*(\mathcal{B}_u)$, as in §1.3, we arrive at

$$Q_{w,q}(u') = \sum_{\substack{i \\ \kappa \in \mathrm{Irr}(A_u)}} \kappa(a(u,u'))\,\mathrm{tr}(wF \mid \mathrm{H}^{2i}(\mathcal{B}_u)_\kappa) \quad \text{for all } u' \in C_u^F. \tag{4.3}$$

Note that the right-hand side only depends on u' via $a(u,u')$.

Suppose that $\chi \in \mathrm{Irr}(W)$ maps to $[u,\kappa] \in \Upsilon$ under the Springer correspondence, and that u is fixed by F. Then

$$Q_{\chi,q}(u') = \kappa(a(u,u'))\sum_i \mathrm{tr}(F \mid \mathrm{H}^{2i}(\mathcal{B}_u)_\kappa) \quad \text{for all } u' \in C_u^F.$$

It follows from the discussion in §4.1 that

$$Q^!_{\chi,q}(u') = \begin{cases} \kappa(a(u,u')) & u' \in C_u^F, \\ 0 & \text{else.} \end{cases}$$

In particular, the values of $Q^!_{\chi,q}$ do not involve q.

As in §1.2, we say that a class $[u,\kappa] \in \Upsilon$, or a representative (u,κ), or just the character κ, is *cuspidal* when it does not arise from $\mathrm{Irr}(W)$ via the Springer correspondence.

Lemma 4.1. *For any $u \in (G^{uni})^F$:*

(1) *If $\kappa \in \mathrm{Irr}(A_u)$ is not cuspidal, then the function on $(G^{uni})^F$ defined by*

$$u' \mapsto \begin{cases} \kappa(a(u,u')) & u' \in C_u^F, \\ 0 & \textit{else} \end{cases}$$

is spanned by the functions $Q_{w,q}$ over $\mathbf{Q}$.

(2) *If there are no cuspidal local systems on C_u, then the function on $(G^{uni})^F$ defined by*

$$u' \mapsto \begin{cases} 1 & u' \text{ is } G^F\text{-conjugate to } u, \\ 0 & \text{else} \end{cases}$$

is spanned by the functions $Q_{w,q}$ over $\mathbf{Q}$.

Proof. (1) follows from the preceding discussion. (2) follows from combining (1) with Schur orthogonality for A_u. □

4.3. **Uniform Class Functions.** Recall the characters $R_w, R_\psi \colon G^F \to \bar{\mathbf{Q}}_\ell$ discussed in Section 2. A $\bar{\mathbf{Q}}_\ell$-valued class function on G^F is called *uniform* when it is spanned by the set $\{R_w\}_w$, or equivalently, $\{R_\psi\}_\psi$.

Lemma 4.2. *For any $w \in W$, the function on G^F defined by*

$$g \mapsto \begin{cases} Q_{w,q}(g) & g \in (G^{uni})^F, \\ 0 & \text{else} \end{cases}$$

is a uniform class function.

As preparation, note that the virtual characters of G^F that we denote by R_w are special cases of Deligne–Lusztig's characters R_T^θ associated with an F-stable maximal torus $T \subseteq G$ and a character $\theta \colon T^F \to \bar{\mathbf{Q}}_\ell^\times$ [DL76]. Namely, again by Galois cohomology and Lang's theorem, G^F-conjugacy classes of F-stable maximal tori of G are in bijection with conjugacy classes of W. We have $R_w = R_T^1$, where 1 denotes the trivial character of T^F, whenever $[w]$ indexes the G^F-conjugacy class of T.

Proof. For any $g \in G^F$ with Jordan decomposition $g = su = us$, with s semisimple and u unipotent, [DL76, Theorem 4.2] states:

$$R_T^\theta(g) = \frac{1}{|(G_s^\circ)^F|} \sum_{\substack{h \in G^F \\ T \subseteq h^{-1} G_s^\circ h}} \theta(h^{-1}sh) Q_T^{G_s^\circ}(u),$$

where G_s° is the neutral component of the centralizer of s in G, and the function $Q_T^{G_s^\circ} \colon (G_s^{\circ,uni})^F \to \bar{\mathbf{Q}}_\ell$ is the restriction to $(G_s^{\circ,uni})^F$ of any Deligne–Lusztig character of $(G_s^\circ)^F$ associated with T. We deduce that

$$\frac{1}{|T^F|} \sum_{\theta \in \mathrm{Irr}(T^F)} R_T^\theta(g) = \begin{cases} Q_T^G(g) & g \text{ is unipotent: } i.e., g = u, \\ 0 & \text{else.} \end{cases}$$

But by Springer's hypothesis, proved in [Kaz77], $Q_w = Q_T^G$ whenever $[w]$ indexes the G^F-conjugacy class of T. □

Corollary 4.3. *For any $u \in (G^{uni})^F$:*

(1) *If $\kappa \in \mathrm{Irr}(A_u)$ is not cuspidal, then the function $\delta_{u,\kappa} \colon G^F \to \mathbf{Q}$ defined by*

$$g \mapsto \begin{cases} \kappa(a(u,g)) & g \in C_u^F, \\ 0 & \text{else} \end{cases}$$

is a uniform class function.

(2) *If there are no cuspidal local systems on C_u, then the function $\delta_u\colon G^F \to \mathbf{Z}$ defined by*

$$g \mapsto \begin{cases} 1 & g \text{ is } G^F\text{-conjugate to } u, \\ 0 & \text{else} \end{cases}$$

is a uniform class function.

Proof. Combine Lemma 4.1 and Lemma 4.2. □

4.4. **Unipotent Lusztig Varieties.** We use Corollary 4.3 to relate our class functions $\tau_{z,\mathsf{v}}\colon W \to \mathbf{Z}[\mathsf{v}^{\pm 1}]$ to the F-fixed points of the Lusztig varieties $Y_{w,u}$ for $u \in (G^{uni})^F$. Recall that explicitly,

$$Y^F_{w,u} = \{B \in \mathcal{B}^F \mid B \xrightarrow{w} u^{-1}Bu\},$$

where $\xrightarrow{w}$ indicates relative position.

Let $(-,-)_W$ be the scalar product on class functions on W. In terms of a fixed square root $q^{1/2}$, let

$$\mathbf{1}_w = q^{\ell(w)/2}\sigma_w \qquad \text{for all } w \in W,$$
$$\psi_q = \psi_{\mathsf{v}}|_{\mathsf{v}\to q^{1/2}} \qquad \text{for all } \psi \in \mathrm{Irr}(W).$$

Moreover, for all $u \in (G^{uni})^F$ and $\kappa \in \mathrm{Irr}(A_u)$, let $spr_{u,\kappa,q}, spr_{u,q}\colon W \to \bar{\mathbf{Q}}_\ell$ be defined by

$$spr_{u,\kappa,q}(w) = \sum_i \mathrm{tr}(wF \mid \mathrm{H}^{2i}(\mathcal{B}_u)_\kappa), \tag{4.4}$$
$$spr_{u,q}(w) := \sum_i \mathrm{tr}(wF \mid \mathrm{H}^{2i}(\mathcal{B}_u)) = Q_{w,q}(u).$$

The following result is inspired by [Tri21].

Theorem 4.4. *Assume that the characteristic of $\mathbf{F}$ is good for G, that its size q is large enough, and that $F\colon G \to G$ defines the split $\mathbf{F}$-structure on G. Let $w \in W$ and $u \in (G^{uni})^F$.*

(1) *If $\kappa \in \mathrm{Irr}(A_u)$ is not cuspidal, then*

$$\frac{1}{|A_u|}\sum_{u' \in C_u^F} \kappa(a(u,u'))|Y^F_{w,u'}| = \sum_{\chi,\psi \in \mathrm{Irr}(W)} \{\chi,\psi\}\psi_q(\mathbf{1}_w)(\chi, spr_{u,\kappa,q})_W, \tag{4.5}$$

(2) *If there are no cuspidal local systems on C_u, then*

$$|Y^F_{w,u}| = \sum_{\chi,\psi \in \mathrm{Irr}(W)} \{\chi,\psi\}\psi_q(\mathbf{1}_w)(\chi, spr_{u,q})_W.$$

Proof. First, by (4.3) and Schur orthogonality, (4.5) can be rewritten as

$$\sum_{u' \in C_u^F} \kappa(a(u'))|Y^F_{w,u'}| = \sum_{\substack{\chi,\psi \in \mathrm{Irr}(W) \\ u' \in C^F}} \kappa(a(u,u'))\{\chi,\psi\}\psi_q(\mathbf{1}_w)(\chi, spr_{u',q})_W.$$

Let $\delta_{u,\kappa}$, δ_u be the class functions on G^F from Corollary 4.3. Let $\mathbf{e}_w, \mathbf{f}_w$ be arbitrary class functions on G^F such that for all $u \in (G^{uni})^F$, we have

$$\begin{aligned}\mathbf{e}_w(u) &= |Y^F_{w,u}|,\\ \mathbf{f}_w(u) &= \sum_{\chi,\psi} \{\chi,\psi\}\psi_q(\mathbf{1}_w)(\chi, spr_{u,q})_W.\end{aligned}$$

(Note that if g is conjugate in G^F to g', then $Y_{w,g} \simeq Y^F_{w,g'}$ and $spr_{g,q} = spr_{g',q}$.) Claims (1) and (2) of the theorem respectively become:

(1) If (u,κ) is not cuspidal, then $(\delta_{u,\kappa}, \mathbf{e}_w)_{G^F} = (\delta_{u,\kappa}, \mathbf{f}_w)_{G^F}$.
(2) If there are no cuspidals on C_u, then $(\delta_u, \mathbf{e}_w)_{G^F} = (\delta_u, \mathbf{f}_w)_{G^F}$.

By Corollary 4.3, it remains to show that the difference between $\mathbf{e}_w$ and $\mathbf{f}_w$ on $(G^{uni})^F$ is the restriction of a function on G^F orthogonal to uniform class functions under $(-,-)_{G^F}$.

As in [Tri21], we use (2.1) to compute

$$|Y^F_{w,g}| = \mathrm{tr}(g \times \mathbf{1}_w \mid R_e) = \sum_{\psi} \rho_\psi(g)\psi_q(\mathbf{1}_w).$$

As in [Lus11a, §1.2], we can expand

$$\rho_\psi = \frac{1}{|W|}\sum_{w\in W}(\rho_\psi, R_w)R_w + \xi_\psi = \sum_{\chi\in \mathrm{Irr}(W)}(\rho_\psi, R_\chi)R_\chi + \xi_\psi,$$

where ξ_ψ is a class function on G^F orthogonal to the uniform class functions. Since $\{-,-\}$ is symmetric [Lus94], we also have $(\rho_\psi, R_\chi) = \{\psi,\chi\} = \{\chi,\psi\}$. Substituting, we arrive at an identity

$$\mathbf{e}_w = \sum_{\chi,\psi}\{\chi,\psi\}\psi_q(\mathbf{1}_w)R_\chi + \Xi_w$$

of class functions, where Ξ_w is orthogonal to the uniform class functions.

Finally, by Springer's hypothesis [Kaz77], $R_w(u) = Q_{w,q}(u) = spr_{u,q}(w)$ for all $u \in (G^{uni})^F$, whence

$$R_\chi(u) = \frac{1}{|W|}\sum_w \chi(w) spr_{u,q}(w) = (\chi, spr_{u,q})_W.$$

Comparing with the definition of $\mathbf{f}_w$, we conclude that $\mathbf{e}_w$ and $\mathbf{f}_w$ differ by Ξ_w on $(G^{uni})^F$, as needed. □

For all $z \in W$, let $\tau_{z,q} = \tau_{z,\mathsf{v}}|_{\mathsf{v}\to q^{1/2}}$. For all $y \le z$, let $p_{y,z}(\mathsf{q}) \in \mathbf{Z}[\mathsf{q}]$ denote the associated Kazhdan–Lusztig polynomial, so that

$$\mathsf{v}^{\ell(w)}c_z = \sum_{y\le z} p_{y,z}(\mathsf{v}^2)\mathbf{1}_y.$$

For instance, if $\ell(z) - \ell(y) \in \{0,1,2\}$, then $p_{y,z}(\mathsf{q}) = 1$. The first nontrivial Kazhdan–Lusztig polynomials occur in type A_3.

Corollary 4.5. *Keep the same assumptions as above on* $\mathbf{F}$, G, F. *Let* $z \in W$ *and* $u \in (G^{uni})^F$.

(1) *If* $\kappa \in \mathrm{Irr}(A_u)$ *is not cuspidal, then*

$$\frac{1}{|A_u|} \sum_{\substack{y \leq z \\ u' \in C_u^F}} p_{y,z}(q)\kappa(a(u,u'))|Y^F_{y,u'}| = q^{\ell(z)/2}(\tau_{z,q}, spr_{u,\kappa,q})_W.$$

(2) *If there are no cuspidal local systems on* C_u, *then*

$$\sum_{y \leq z} p_{y,z}(q)|Y^F_{y,u}| = q^{\ell(z)/2}(\tau_{z,q}, spr_{u,q})_W.$$

Proof. By the definitions of $p_{y,z}$ and $\tau_{z,\mathsf{v}}$. □

Corollary 4.6. *Keep the same assumptions as above on* $\mathbf{F}$, G, F. *Let* $z \in W$ *and* $u \in (G^{uni})^F$. *Then the Laurent polynomials* $\alpha_z^\psi(\mathsf{v})$ *defined by* (1.3) *satisfy the following constraints:*

(1) *If* $\kappa \in \mathrm{Irr}(A_u)$ *is not cuspidal, we have*

$$\sum_{\substack{y \leq z \\ u' \in C_u^F}} p_{y,z}(q)\kappa(a(u,u'))|Y^F_{y,u'}| = q^{\ell(z)/2} \sum_{\psi \in \mathrm{Irr}(W)} \alpha_z^\psi(q^{1/2})(spr_\psi, spr_{u,\kappa,q})_W. \tag{4.6}$$

(2) *If there are no cuspidal local systems on* C_u, *then*

$$\sum_{y \leq z} p_{y,z}(q)|Y^F_{y,u}| = q^{\ell(z)/2} \sum_{\psi \in \mathrm{Irr}(W)} \alpha_z^\psi(q^{1/2})(spr_\psi, spr_{u,q})_W. \tag{4.7}$$

Proof. By substituting into the preceding corollary. □

Corollary 4.6 expresses the $\alpha_z^\psi(\mathsf{v})$ solely in terms of objects over the unipotent locus G^{uni}. Another reason it seems worth stating is that—to us—the point counts $|Y^F_{y,u}|$ seem more concrete than either the exotic Fourier transform in Section 2 or the intersection cohomologies in Section 3.

4.5. **Split Representatives.** One might ask whether, as we run over non-cuspidal pairs (u, κ) and large q, the constraints (4.6) determine the $\alpha_z^\psi(\mathsf{v})$ uniquely. To answer this, we need to analyze the scalar products $(spr_\psi, spr_{u,\kappa,q})$. Note that spr_ψ is defined in terms of Springer fibers over the complex numbers, while $spr_{u,\kappa,q}$ is defined in terms of Springer fibers over a finite field.

Naïvely, if the Springer correspondence sends $\psi \mapsto [u, \kappa]$, and $u \in (G^{uni})^F$, then spr_ψ should be the "$F \to \mathrm{Id}$" limit of $spr_{u,\kappa,q}$, according to the definitions in (1.2) and (4.4). To make this idea rigorous, we might try to lift $spr_{u,\kappa,q}$ to a function $spr_{u,\kappa,\mathsf{q}}$ in which the number q is replaced by an indeterminate q. Then we would also need a precise comparison between the complex and finite-field incarnations of Springer theory.

Fix an F-stable unipotent conjugacy class $C \subseteq G$. As explained in [Som11, §2.2], it is almost always possible to choose an element $u \in C^F$ for which F acts trivially

on the top-degree cohomology $\mathrm{H}^{2d_u}(\mathcal{B}_u)$. Such an element u is said to be *split*. It is unique in C^F up to G^F-conjugacy. If G is almost-simple, then the only situation where split elements do not exist occurs when G is of type E_8 and $q \equiv -1 \pmod 3$, and then only for a single C, labeled $D_8(a_3)$ in [Car93, BS84]. Here, for any $u \in C^F$, it turns out that $A_u \simeq S_3$, which has 3 conjugacy classes, and C^F also decomposes into 3 conjugacy classes of G^F. For general G, we refer to such classes $C \subseteq G$ arising from factors of type E_8 as *exceptional classes*.

For any $u \in (G^{uni})^F$ and $\kappa \in \mathrm{Irr}(A_u)$, let $spr_{u,\kappa,\mathsf{q}}, spr_{u,\mathsf{q}} \colon W \to \mathbf{C}[\mathsf{q}]$ be defined by

$$spr_{u,\kappa,\mathsf{q}}(w) = \sum_i \mathsf{q}^i \operatorname{tr}(w \mid \mathrm{H}^{2i}(\mathcal{B}_u)_\kappa),$$
$$spr_{u,\mathsf{q}}(w) = \sum_i \mathsf{q}^i \operatorname{tr}(w \mid \mathrm{H}^{2i}(\mathcal{B}_u)).$$

It is explained in [BS84, §5(C)] that for a split element u and any integer i, the F-action on $\mathrm{H}^{2i}(\mathcal{B}_u)$ is simply multiplication by q^i. Therefore, for split u, we have $spr_{u,\kappa,\mathsf{q}}|_{\mathsf{t}\to q} = spr_{u,\kappa,q}$ and $spr_{u,\mathsf{q}}|_{\mathsf{q}\to q} = spr_{u,q}$.

Next, let $G_{\mathbf{C}}$ be the connected reductive algebraic group over $\mathbf{C}$ with the same root datum as G. Recall that conjugacy classes of $G_{\mathbf{C}}$ are in bijection with F-stable conjugacy classes of G. Let $C_{\mathbf{C}} \subseteq G_{\mathbf{C}}$ be the unipotent conjugacy class corresponding to $C \subseteq G$. As we vary $\tilde{u} \in C_{\mathbf{C}}(\mathbf{C})$, the graded character of the representation of W on $\mathrm{H}^*(\mathcal{B}_{\tilde{u}})$ does not depend on $\tilde{u}$. As long as C is not exceptional, we can pick $\tilde{u}$ to be the lift of a split element $u \in C^F$ to $C_{\mathbf{C}}(R)$, for some finitely generated subring $R \subseteq \mathbf{C}$ and prime $\mathfrak{p} \subseteq R$ such that $R/\mathfrak{p} = \mathbf{F}$. In this situation, $A_{\tilde{u}} \simeq A_u$, allowing us to regard $\kappa \in \mathrm{Irr}(A_u)$ as an element $\tilde{\kappa} \in \mathrm{Irr}(A_{\tilde{u}})$. We then have

$$spr_{u,\kappa,\mathsf{q}}(w) = \sum_i \mathsf{q}^i \operatorname{tr}(w \mid \mathrm{H}^{2i}(\mathcal{B}_{\tilde{u}})_{\tilde{\kappa}}),$$
$$spr_{u,\mathsf{q}}(w) = \sum_i \mathsf{q}^i \operatorname{tr}(w \mid \mathrm{H}^{2i}(\mathcal{B}_{\tilde{u}})).$$

In particular: $\mathrm{H}^{2d_u}(\mathcal{B}_u)_\kappa$ affords ψ if and only if $\mathrm{H}^{2d_{\tilde{u}}}(\mathcal{B}_{\tilde{u}})_{\tilde{\kappa}}$ does, and in this case, we have $spr_{u,\kappa,\mathsf{q}}|_{\mathsf{q}\to 1} = spr_\psi$.

In what follows, let Υ_{spl} be a full and irredundant set of representatives (u,κ) for each class $[u,\kappa] \in \Upsilon$, such that if C_u is not exceptional, then we choose u to be split.

Lemma 4.7. *The matrix of scalar products $((spr_\psi, spr_{u,\kappa,\mathsf{q}})_W)_{\psi,[u,\kappa]}$, indexed by $\psi \in \mathrm{Irr}(W)$ and $[u,\kappa] \in \Upsilon_{spl}$, is invertible.*

Proof. We can reduce to the case where W is irreducible, and hence, to the case where G is almost-simple. Then the exceptional conjugacy classes in type E_8 can be handled separately, using [BS84].

So suppose that G admits no exceptional F-stable unipotent conjugacy classes. By the discussion above the lemma, the matrix $((spr_\psi, spr_{u,\kappa,\mathsf{q}})_W)_{\psi,[u,\kappa]}$ specializes at $\mathsf{q} \to 1$ to the matrix $((spr_\psi, spr_\chi)_W)_{\psi,\chi}$, where ψ, χ both run over $\mathrm{Irr}(W)$. But the linear map on class functions that sends $\psi \mapsto spr_\psi$ is invertible (*e.g.*, by Lusztig–Shoji [Lus86, Sho88]), and the matrix $((\psi,\chi)_W)_{\psi,\chi}$ is simply the identity matrix. $\square$

Corollary 4.8. *Keep the same assumptions as in Corollary 4.6. If q is large enough, then as we run over $[u,\kappa] \in \Upsilon_{spl}$, the constraints (4.6) determine the values $\alpha_z^\psi(q^{1/2})$ uniquely. Hence, as we also run over infinitely many choices of $\mathbf{F}$, they determine the Laurent polynomials $\alpha_z^\psi(\mathsf{v})$ uniquely.*

4.6. **Fake Degrees and Kazhdan–Lusztig Polynomials.** To conclude this section, we examine a special case of Corollary 4.6 more carefully.

Writing 1 for the identity of G, we see that $Y_{e,1} = \mathcal{B}$, the flag variety of G, whereas $Y_{y,1} = \emptyset$ for all $y \neq e$. Hence, for $u = 1$, the left-hand side of (4.7) becomes $p_{e,z}(q)|\mathcal{B}^F|$. Here, by Bruhat, we know that $|\mathcal{B}^F| = P_W(q)$, where $P_W(\mathsf{q}) \in \mathbf{Z}[\mathsf{q}]$ is the polynomial defined by

$$P_W(\mathsf{q}) = \sum_{w \in W} \mathsf{q}^{\ell(w)}.$$

By Borel, we also know that $spr_{\{1\},\mathsf{q}}$ is the graded character of the so-called module of polynomial coinvariants of the reflection representation of W. Explicitly,

$$spr_{\{1\},\mathsf{q}} = \sum_{\chi \in \mathrm{Irr}(W)} P_\chi(\mathsf{q})\chi,$$

where $P_\chi(\mathsf{q}) \in \mathbf{Z}[\mathsf{q}]$ is the *fake degree* polynomial for χ [Car93, §11.3], essentially determined by the identity $P_\chi(q) = R_\chi(1)$.

There are no cuspidal local systems on the conjugacy class $\{1\}$. So altogether, Corollary 4.6(2) yields the following formula for the Kazhdan–Lusztig polynomial $p_{e,z}$, once we observe that it holds at infinitely many choices of $\mathbf{F}$.

Corollary 4.9. *For any $z \in W$, we have*

$$p_{e,z}(\mathsf{v}^2) = \frac{1}{P_W(\mathsf{v}^2)} \sum_{\chi,\psi \in \mathrm{Irr}(W)} P_\chi(\mathsf{v}^2)\alpha_z^\psi(\mathsf{v})(\chi, spr_\psi)_W. \tag{4.8}$$

Remark 4.10. By taking $W' = \{e\}$ in (3.3), we see that

$$\tau_{e,\mathsf{v}} = \sum_{\chi,\psi} \{\chi,\psi\}\psi(1)\chi = \sum_\psi \psi(1)\psi,$$

the character of the regular representation of W. In our conventions for the Springer action, this is also spr_ε, where ε is the sign character of W. So the $z = e$ case of Conjecture D holds, with $\alpha_e^\varepsilon(\mathsf{v}) = 1$ and $\alpha_e^\psi(\mathsf{v}) = 0$ for $\psi \neq \varepsilon$.

For $z = e$, the left-hand side of (4.8) is just $p_{e,e}(\mathsf{v}^2) = 1$, while the sum on the right-hand side simplifies to $\sum_\chi P_\chi(\mathsf{v}^2)\chi(1)$. We thereby recover the identity $P_W(\mathsf{q}) = \sum_\chi P_\chi(\mathsf{q})\chi(1)$.

5. Examples

Recall that in [Hai93], Haiman verified Conjecture A up through rank 6. So to prove Theorem 1.4—that is, to verify Conjecture D for all Weyl groups of rank ≤ 5—it remains to verify it for the irreducible Weyl groups of types BC_2, G_2, BC_3, BC_4, D_4, F_4, D_5, BC_5.

5.1. **Methodology.** We can rewrite identity (1.3), defining the Laurent polynomials $\alpha_z^\psi(\mathsf{v})$, as the collection of identities

$$(\chi, \tau_{z,\mathsf{v}})_W = \sum_{\psi \in \mathrm{Irr}(W)} \alpha_z^\psi(\mathsf{v})(\chi, spr_\psi)_W \quad \text{for } \chi \in \mathrm{Irr}(W). \tag{5.1}$$

Recall that the left-hand side is $\sum_\psi \{\chi, \psi\}\psi_{\mathsf{v}}(c_z)$. In CHEVIE, it can be computed using `HeckeCharValues`, `UnipotentCharacter`, and `AlmostCharacter`. The families $\mathcal{F} \subseteq \mathrm{Irr}(W)$, and hence the Laurent polynomials $\alpha_z^{\mathcal{F}}(\mathsf{v})$ defined by (1.4), can then be obtained from `UnipotentCharacters`.

What makes (5.1) convenient is that $(\chi, spr_\psi)_W = (Q_\chi^!, Q_\psi)_{G^F}$, in the notation of Section 4. The latter scalar products are well-attested in the literature on the Lusztig–Shoji algorithm, where $(Q_\chi^!, Q_\psi)_{G^F}$ is usually denoted $P_{\psi,\chi}$: See [GM99, Ach21, Ach09]. These values, which may be viewed as generalized Kostka numbers, are stored in CHEVIE in the function `ICCTable`.

To actually calculate the $\alpha_z^\psi(\mathsf{v})$, we used data from CHEVIE and Python scripts written with help from Claude Sonnet 4.6. To calculate the standard Coxeter elements c and the respective lists of c-sortable elements, we used a script written in Sage. All of these data and scripts are available at:

https://mqtrinh.github.io/math/research/code/chevie/

The rest of this section summarizes the results type by type.

5.2. **Notation.** In the tables for $(\chi, spr_\psi)_W$, we list χ along rows and ψ along columns. We label spr_ψ by the class $[u, \kappa]$ such that $\psi \mapsto [u, \kappa]$ under Springer. We use the labels for χ and $[u, \kappa]$ native to CHEVIE, in turn based upon [Car93], except that we write ε in place of $[1, 1_{A_1}]$ to make the tables less wide.

In the tables for $(\chi, \tau_{z,\mathsf{v}})_W$, we list χ along rows and z along columns. We write $w_\circ$ for the longest element of W. To compactify the table entries, we write

$$(\mathsf{a}_m \cdots \mathsf{a}_1 \mathsf{a}_0) := \mathsf{a}_0 + \sum_{i=1}^{m} \mathsf{a}_i(\mathsf{v}^i + \mathsf{v}^{-i}) \quad \text{for any positive integers } \mathsf{a}_0, \mathsf{a}_1, \ldots, \mathsf{a}_m.$$

We obtain the values $\alpha_z^\psi(\mathsf{v})$ by inverting the matrix $((\chi, spr_\psi)_W)_{\chi,\psi}$, then applying it on the left to the matrix $((\chi, \tau_{z,\mathsf{v}})_W)_{\chi,z}$.

If ψ is the special character of some family, then we write ψ or $\mathcal{F}_\psi$ to label the family. When discussing the $\alpha_z^\psi(\mathsf{v})$, we mark the character of the reflection representation with $\star$, and its Lusztig–Spaltenstein dual with $\star\star$ [Car93, §13.4].

For any z and $\bullet \in \{\psi, \mathcal{F}\}$, we use the following notation for the properties we must check:

- P For all $z \in \Sigma$, the nonzero coefficients of $\alpha_z^\bullet$ are all positive.
- $\mathsf{P}_\pm$ For all $z \in \Sigma$, the nonzero coefficients of $\alpha_z^\bullet$ are either all positive or all negative.
- U For all $z \in \Sigma$, the nonzero coefficients of $\alpha_z^\bullet$ are unimodal in absolute value.

5.3. **Type BC_2.** The table of values $(\chi, spr_\psi)_W$:

	5	$311^{(11)}$	311	221	ε
2.	1		1	1	1
11.		1			1
1.1			1	1	2
.2				1	1
.11					1

We write s, t for the simple reflections respectively numbered $1, 2$ in CHEVIE. The table of Laurent polynomials $(\chi, \tau_{z,\mathsf{v}})_W$:

	e	t	s	ts, st	tst	sts	$w_\circ$
2.	1	(10)	(10)	(102)	(1020)	(1020)	(10202)
11.	1		(10)	1	(10)		
1.1	2	(10)	(10)	1			
.2	1	(10)		1		(10)	
.11	1						

The table of Laurent polynomials $\alpha_z^\psi(\mathsf{v})$:

	e	t	s	ts, st	tst	sts	$w_\circ$
2.				(101)	(1020)	(1020)	(10202)
11.			(10)	1	(10)		
$\star$1.1			(10)			$-(10)$	
.2		(10)		1		(10)	
.11	1						

The families of $\mathrm{Irr}(W)$ are

$$\mathcal{F}_{2.} = \{2.\}, \quad \mathcal{F}_{1.1} = \{11., 1.1, .2\}, \quad \mathcal{F}_{.11} = \{.11\}.$$

The table of Laurent polynomials $\alpha_z^{\mathcal{F}}(\mathsf{v})$:

	e	t	s	ts, st	tst	sts	$w_\circ$
2.				(101)	(1020)	(1020)	(10202)
1.1		(10)	(20)	2	(10)		
.11	1						

So Conjecture C holds.

5.4. **Type G_2.** The table of $(\chi, spr_\psi)_W$:

	G_2	$G_2(a_1)^{(21)}$	$G_2(a_1)$	$\tilde{A}_1$	A_1	ε
$\phi_{1,0}$	1		1	1	1	1
$\phi'_{1,3}$		1		1		1
$\phi_{2,1}$			1	1	1	2
$\phi_{2,2}$				1	1	2
$\phi''_{1,3}$					1	1
$\phi_{1,6}$						1

We write s, t for the simple reflections respectively numbered $1, 2$ in CHEVIE. The table of $(\chi, \tau_{z,\mathsf{v}})_W$:

	e	t	s	ts, st	tst	sts	$tsts, stst$	$tstst$	$ststs$	$w_\circ$
$\phi_{1,0}$	1	(10)	(10)	(102)	(1020)	(1020)	(10202)	(102020)	(102020)	(1020202)
$\phi'_{1,3}$	1	(10)		1		(10)	1	(10)		
$\phi_{2,1}$	2	(10)	(10)	1		(10)				
$\phi_{2,2}$	2	(10)	(10)	2	(10)		1			
$\phi''_{1,3}$	1		(10)	1	(10)		1		(10)	
$\phi_{1,6}$	1									

The table of $\alpha_z^\psi(\mathsf{v})$:

	e	t	s	ts, st	tst	sts	$tsts, stst$	$tstst$	$ststs$	$w_\circ$
$\phi_{1,0}$				(101)	(1020)	(1020)	(10202)	(102020)	(102020)	(1020202)
$\phi'_{1,3}$							1	(10)	(10)	
$\star\phi_{2,1}$				-1	$-(10)$	$-(10)$	-1			
$\phi_{2,2}$		(10)		1		(10)			$-(10)$	
$\phi''_{1,3}$			(10)	1	(10)		1		(10)	
$\phi_{1,6}$	1									

The families of $\mathrm{Irr}(W)$ are

$$\mathcal{F}_{\phi_{1,0}} = \{\phi_{1,0}\}, \quad \mathcal{F}_{\phi_{2,1}} = \{\phi'_{1,3}, \phi_{2,1}, \phi_{2,2}, \phi''_{1,3}\}, \quad \mathcal{F}_{\phi_{1,6}} = \{\phi_{1,6}\}.$$

The table of $\alpha_z^{\mathcal{F}}(\mathsf{v})$:

	e	t	s	ts, st	tst	sts	$tsts, stst$	$tstst$	$ststs$	$w_\circ$
$\phi_{1,0}$				(101)	(1020)	(1020)	(10202)	(102020)	(102020)	(1020202)
$\phi_{2,1}$		(10)	(10)	1			1	(10)	(10)	
$\phi_{1,6}$	1									

So Conjecture D holds. We note that every element of W is c-sortable for some standard Coxeter element c.

5.5. **Type BC_3.** We omit the tables of $(\chi, spr_\psi)_W$ and $(\chi, \tau_{z,\mathsf{v}})_W$. We write s, t, u for the simple reflections respectively numbered $1, 2, 3$ in CHEVIE, so that $(st)^4 =$

$(tu)^3 = e$. The tables of $\alpha_z^\psi(\mathsf{v})$:

	e	u,t	s	ut,tu	ts,st	su	tst	uts,tsu,sut,stu	tut	sts	$utst,tstu$
3.								(1010)			(10202)
21.					(101)		(1020)	(10)		(1020)	(203)
2.1					(101)		(1020)			(1020)	(101)
.3				(101)				(10)	(1020)		1
1.2				(101)				(10)	(1020)		1
11.1						(102)		(10)			
111.			(10)				(10)				1
⋆⋆1.11			(10)							−(10)	
.21		(10)		1						(10)	
.111	1										

	$tuts,suts,stsu,stut$	$tsut$	$stst$	$utstu$	$tstut,tutst$	$tsuts,stsut$	$sutst,ststu$
3.	(10202)	(10202)		(103040)	(103040)	(103030)	(102020)
21.	(102)	(102)	(10202)	(2050)	(1030)	(1030)	(1020)
21.			(10202)	(1020)		(1020)	(1020)
.3	(102)	(102)		(10)	(10)	(20)	(10)
1.2	(102)	(102)		(10)	(10)	(10)	
11.1		(102)				(10)	
111.				(10)			
⋆⋆1.11							
.21							
.111							

	$stuts$	$tutstu$	$tstuts,stutst$	$tsutst,ststut$	$sutstu$	$stsuts$	$tstutst$
3.	(103040)	(1030506)	(1040606)	(1030404)	(1030404)	(1030404)	(10407080)
21.	(1030)	(10304)	(204)	(102)	(10304)	(102)	(1030)
21.			(102)	(102)	(10304)	(102)	(1030)
.3	(1030)		(204)	(102)	(102)	(102)	(2050)
1.2	(1030)		(102)				(1020)
11.1						(102)	
111.							
⋮							

	$tsutstu,$ $stutstu$	$ststuts,$ $stsutst$	$tstutstu$	$ststutst$	$stsutstu$	$w_\circ$
3.	(10407080)	(10305060)	(103050606)	(103050708)	(104080[11]0[12])	(1030507080)
21.	(1030)	(10)			(102)	
21.	(1030)	(10)	(10304)		(102)	
.3	(1030)	(1020)		(10202)	(10304)	
1.2			(10304)			
11.1						
111.						
⋮						

The families of $\mathrm{Irr}(W)$ are

$$\mathcal{F}_{3.} = \{3.\}, \qquad \mathcal{F}_{2.1} = \{21., 2.1, .3\}, \qquad \mathcal{F}_{1.2} = \{1.2\},$$
$$\mathcal{F}_{11.1} = \{11.1\}, \quad \mathcal{F}_{1.11} = \{111., 1.11, .21\}, \quad \mathcal{F}_{.111} = \{.111\}.$$

We omit the tables of $\alpha_z^{\mathcal{F}}(\mathsf{v})$. Properties P and U hold for all $z \in W$ and families $\mathcal{F}$, so Conjecture C holds.

5.6. **Type BC_4.** We omit the tables. The elements of $\mathrm{Irr}(W)$ are labeled by the following bipartitions of 4:

$$\begin{array}{cccccccccc} 4., & 31., & 3.1, & .4, & 22., & 2.2, & 1.3, & 21.1, & 211., & 2.11, \\ 11.2, & .31, & 1.21, & 111.1, & 11.11, & .22, & 1111., & 1.111, & .211, & .1111. \end{array}$$

The table of cases where P, $\mathsf{P}_\pm$, U fails for some $z \in W$:

	ψ	$\mathcal{F}$
P	$2.2, 11.11, \star\star 1.111$	*none*
$\mathsf{P}_\pm$	*none*	*none*
U	*none*	*none*

So Conjecture C holds.

5.7. **Type D_4.** We omit the tables. The elements of $\mathrm{Irr}(W)$ are labeled as follows:

$$\begin{array}{ccccccc} .4, & 1.3, & .31, & 2-, & 2+, & 11.2, & 1.21, \\ .22, & .211, & 11-, & 11+, & 1.111, & .1111 & \end{array}$$

The table of cases where P, $\mathsf{P}_\pm$, U fails for some $z \in W$:

	ψ	$\mathcal{F}$
P	$\star 1.3, 1.21$	$\mathcal{F}_{1.3}$
$\mathsf{P}_\pm$	*none*	*none*
U	*none*	*none*

We write s_i for the simple reflection numbered i in CHEVIE, so that $(s_1s_3)^3 = (s_2s_3)^3 = (s_3s_4)^3 = e$. There are 8 standard Coxeter elements c:

c	P holds for all c-sortable z and $\mathcal{F}$
$s_1s_3s_2s_4$	
$s_2s_4s_3s_1$	
$s_3s_2s_1s_4$	✓
$s_2s_3s_1s_4$	
$s_1s_4s_3s_2$	
$s_4s_3s_2s_1$	✓
$s_2s_1s_3s_4$	✓
$s_2s_1s_4s_3$	✓

So Conjecture D holds.

5.8. **Type F_4.** We omit the tables. The elements of $\mathrm{Irr}(W)$ are labeled as follows:

$$\begin{array}{ccccccccc} \phi_{1,0}, & \phi'_{2,4}, & \phi_{4,1}, & \phi''_{2,4}, & \phi_{9,2}, & \phi'_{8,3}, & \phi''_{8,3}, & \phi'_{1,12}, & \phi''_{6,6}, \\ \phi'_{9,6}, & \phi_{12,4}, & \phi'_{4,7}, & \phi_{16,5}, & \phi'_{6,6}, & \phi_{4,8}, & \phi''_{9,6}, & \phi''_{4,7}, & \phi'_{8,9}, \\ \phi''_{1,12}, & \phi''_{8,9}, & \phi_{9,10}, & \phi'_{2,16}, & \phi_{4,13}, & \phi''_{2,16}, & \phi_{1,24}. & & \end{array}$$

The table of cases where P, $\mathsf{P}_\pm$, U fails for some $z \in W$:

	ψ	$\mathcal{F}$
P	$\star\phi_{4,1}, \phi_{9,2}, \phi'_{9,6}, \phi_{12,4}, \phi'_{4,7}, \phi_{16,5}, \phi'_{6,6}, \phi''_{9,6}, \phi''_{4,7}, \star\star\phi_{4,13}$	$\mathcal{F}_{\phi_{4,1}}, \mathcal{F}_{\phi_{9,2}}, \mathcal{F}_{\phi_{12,4}}$
$\mathsf{P}_\pm$	$\star\phi_{4,1}, \phi_{9,2}, \phi_{12,4}$	$\mathcal{F}_{\phi_{4,1}}, \mathcal{F}_{\phi_{9,2}}$
U	$\star\phi_{4,1}, \phi_{9,2}$	$\mathcal{F}_{\phi_{4,1}}, \mathcal{F}_{\phi_{9,2}}$

We note that $\star\phi_{4,1}$ $\phi_{9,2}$, $\phi_{12,4}$ are all special characters. Under Springer, they respectively map to the three distinguished unipotent conjugacy classes in the group G, labeled $F_4(a_1)$, $F_4(a_2)$, $F_4(a_3)$ in [Car93]. We also note that if $\psi = \phi_{9,2}$, then U only fails for a single z: namely,

$$z = s_2s_3s_2s_1s_3s_2s_3s_4s_3s_2s_1s_3s_2s_4s_3s_2,$$

where s_i is the reflection numbered i in CHEVIE.

We write s_i for the simple reflection numbered i in CHEVIE, so that $(s_1s_2)^3 = (s_2s_3)^4 = (s_3s_4)^3 = e$. There are 8 standard Coxeter elements c:

c	P	$\mathsf{P}_\pm$	U	...holds for all c-sortable z and $\mathcal{F}$
$s_3s_4s_1s_2$	✓	✓	✓	
$s_2s_3s_4s_1$	✓	✓	✓	
$s_1s_2s_3s_4$		✓	✓	
$s_3s_4s_2s_1$	✓	✓	✓	
$s_4s_3s_1s_2$	✓	✓	✓	
$s_4s_1s_2s_3$		✓	✓	
$s_4s_2s_3s_1$	✓	✓	✓	
$4s_3s_2s_1$		✓	✓	

So Conjecture D holds.

5.9. **Type BC_5.** We omit the tables. The elements of $\mathrm{Irr}(W)$ are labeled by the following bipartitions of 5:

5.,	41.,	4.1,	.5,	32.,	3.2,	1.4,	2.3,	31.1,
311.,	3.11,	22.1,	21.2,	221.,	.41,	2.21,	11.3,	1.31,
211.1,	21.11,	111.2,	11.21,	.32,	1.22,	2111.,	2.111,	.311,
1.211,	111.11,	1111.1,	11.111,	.221,	11111.,	1.1111,	.2111,	.11111

The table of cases where P, $\mathsf{P}_\pm$, U fails for some $z \in W$:

	ψ	$\mathcal{F}$
P	$3.2, 21.2, 2.21, 11.21, 11.111, \star\star 1.1111$	*none*
$\mathsf{P}_\pm$	3.2	*none*
U	3.2	*none*

So Conjecture C holds.

We note that this is the first case where U fails for some $z \in W$ and $\psi \in \mathrm{Irr}(W)$ such that z is c-sortable for some z. Namely, it fails for

$$z = s_2s_1s_3s_2s_1s_4s_3s_2s_1s_5s_4s_3s_2s_1s_2s_3 \quad \text{and} \quad \psi = 3.2,$$

where s_i is the simple reflection numbered i in CHEVIE. This choice of z is c-sortable for $c = s_2s_1s_3s_5s_4$ alone. For this z and ψ, the nonzero coefficients of $\alpha_z^\psi(\mathsf{v})$ are all negative, so $\mathsf{P}_\pm$ still holds.

In all other types discussed in this section, if z is c-sortable for some c, then $\mathsf{P}_\pm$ and U hold for all ψ.

5.10. **Type D_5.** We omit the tables. The elements of $\mathrm{Irr}(W)$ are labeled as follows:

.5,	1.4,	.41,	2.3,	11.3,	1.31,	2.21,	.32	1.22,
.311,	11.21,	111.2,	1.211,	.221,	11.111	.2111,	1.1111,	.11111

The table of cases where P, $\mathsf{P}_\pm$, U fails for some $z \in W$:

	ψ	$\mathcal{F}$
P	$\star 1.4, 1.211, 11.3, 1.31$	$1.4, 1.31$
$\mathsf{P}_\pm$	$1.4, 1.31$	1.4
U	$1.4, 1.31$	1.4

We write s_i for the simple reflection numbered i in CHEVIE, so that $(s_1s_3)^3 = (s_2s_3)^3 = (s_3s_4)^3 = (s_4s_5)^3 = e$. There are 16 standard Coxeter elements c:

c	P	$\mathsf{P}_\pm$	U	...holds for all c-sortable z and $\mathcal{F}$
$s_2s_5s_4s_3s_1$		✓	✓	
$s_2s_1s_4s_3s_5$	✓	✓	✓	
$s_1s_3s_2s_5s_4$		✓	✓	
$s_2s_4s_3s_1s_5$		✓	✓	
$s_1s_5s_4s_3s_2$		✓	✓	
$s_2s_3s_1s_5s_4$		✓	✓	
$s_4s_3s_2s_1s_5$	✓	✓	✓	
$s_1s_3s_2s_4s_5$		✓	✓	
$s_2s_3s_1s_4s_5$		✓	✓	
$s_2s_1s_3s_5s_4$	✓	✓	✓	
$s_3s_2s_1s_5s_4$	✓	✓	✓	
$s_2s_1s_5s_4s_3$	✓	✓	✓	
$s_5s_4s_3s_2s_1$	✓	✓	✓	
$s_1s_4s_3s_2s_5$		✓	✓	
$s_3s_2s_1s_4s_5$	✓	✓	✓	
$s_2s_1s_3s_4s_5$	✓	✓	✓	

So Conjecture D holds.

References

[AA08] P. N. Achar and A.-M. Aubert. Springer correspondences for dihedral groups. *Transform. Groups*, 13(1):1–24, 2008. doi:10.1007/s00031-008-9004-2.

[Ach09] Pramod N. Achar. Springer theory for complex reflection groups. In *RIMS Kôkyûroku 1647. Expansion of Combinatorial Representation Theory. Ed. Hyohe Miyachi*, pages 97–112. Research Institute for Mathematical Sciences, 2009. URL: https://www.kurims.kyoto-u.ac.jp/~kyodo/kokyuroku/contents/1647.html.

[Ach21] Pramod N. Achar. *Perverse sheaves and applications to representation theory*, volume 258 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, [2021] ©2021. doi:10.1090/surv/258.

[AN25] Alex Abreu and Antonio Nigro. A geometric approach to characters of Hecke algebras. *J. Reine Angew. Math.*, 821:53–114, 2025. doi:10.1515/crelle-2024-0098.

[BC18] Patrick Brosnan and Timothy Y. Chow. Unit interval orders and the dot action on the cohomology of regular semisimple Hessenberg varieties. *Adv. Math.*, 329:955–1001, 2018. doi:10.1016/j.aim.2018.02.020.

[BHL25] Patrick Brosnan, Jaehyun Hong, and Donggun Lee. Geometry of regular semisimple Lusztig varieties, 2025. v1. arXiv:2504.15868.

[BS84] W. M. Beynon and N. Spaltenstein. Green functions of finite Chevalley groups of type E_n $(n = 6, 7, 8)$. *J. Algebra*, 88:584–614, 1984. doi:10.1016/0021-8693(84)90084-X.

[Car93] Roger W. Carter. *Finite groups of Lie type*. Wiley Classics Library. John Wiley & Sons, Ltd., Chichester, 1993. Conjugacy classes and complex characters, Reprint of the 1985 original, A Wiley-Interscience Publication.

[CHSS16] Samuel Clearman, Matthew Hyatt, Brittany Shelton, and Mark Skandera. Evaluations of Hecke algebra traces at Kazhdan-Lusztig basis elements. *Electron. J. Comb.*, 23(2):research paper p2.7, 56, 2016. URL: www.combinatorics.org/ojs/index.php/eljc/article/view/v23i2p7.

[Del77] Pierre Deligne. Théoremes de finitude en cohomologie ℓ-adique (avec un appendice par L. Illusie). Semin. Geom. algebr. Bois-Marie, SGA 4 1/2, Lect. Notes Math. 569, 233-261, 1977. doi:10.1007/bfb0091524.

[DL76] Pierre Deligne and G. Lusztig. Representations of reductive groups over finite fields. *Ann. Math. (2)*, 103:103–161, 1976. doi:10.2307/1971021.

[GHL+96] Meinolf Geck, Gerhard Hiss, Franz Lübeck, Gunter Malle, and Götz Pfeiffer. CHEVIE – A system for computing and processing generic character tables for finite groups of Lie type, Weyl groups and Hecke algebras. *Appl. Algebra Engrg. Comm. Comput.*, 7:175–210, 1996.

[GM99] Meinolf Geck and Gunter Malle. On special pieces in the unipotent variety. *Exp. Math.*, 8(3):281–290, 1999. doi:10.1080/10586458.1999.10504405.

[GMR+25] Sean T. Griffin, Anton Mellit, Marino Romero, Kevin Weigl, and Joshua Jeishing Wen. On Macdonald expansions of q-chromatic symmetric functions and the Stanley-Stembridge Conjecture, 2025. v1. arXiv:2504.06936.

[Hai93] Mark Haiman. Hecke algebra characters and immanant conjectures. *J. Am. Math. Soc.*, 6(3):569–595, 1993. doi:10.2307/2152777.

[He25] Xuhua He. On affine Lusztig varieties. *Ann. Sci. Éc. Norm. Supér. (4)*, 58(3):749–776, 2025. doi:10.24033/asens.2615.

[Hik25] Tatsuyuki Hikita. A proof of the Stanley–Stembridge conjecture, 2025. v2. arXiv:2410.12758.

[Kaz77] David Kazhdan. Proof of Springer's hypothesis. *Isr. J. Math.*, 28:272–286, 1977. doi:10.1007/BF02760635.

[Kim18] Dongkwan Kim. On total Springer representations for classical types. *Sel. Math., New Ser.*, 24(5):4141–4196, 2018. doi:10.1007/s00029-018-0438-7.

[KL79] David Kazhdan and George Lusztig. Representations of Coxeter groups and Hecke algebras. *Invent. Math.*, 53:165–184, 1979. doi:10.1007/BF01390031.

[KV12] Robert Kottwitz and Eva Viehmann. Generalized affine Springer fibres. *J. Inst. Math. Jussieu*, 11(3):569–609, 2012. doi:10.1017/S147474801100020X.

[Lus84a] G. Lusztig. Intersection cohomology complexes on a reductive group. *Invent. Math.*, 75(2):205–272, 1984. doi:10.1007/BF01388564.

[Lus84b] George Lusztig. *Characters of reductive groups over a finite field*, volume 107 of *Annals of Mathematics Studies*. Princeton University Press, Princeton, NJ, 1984. doi:10.1515/9781400881772.

[Lus85a] George Lusztig. Character sheaves. II. *Adv. Math.*, 57:226–265, 1985. doi:10.1016/0001-8708(85)90064-7.

[Lus85b] George Lusztig. Character sheaves. III. *Adv. Math.*, 57:266–315, 1985. doi:10.1016/0001-8708(85)90065-9.

[Lus86] George Lusztig. Character sheaves. V. *Adv. Math.*, 61:103–155, 1986. doi:10.1016/0001-8708(86)90071-X.

[Lus93] George Lusztig. Appendix: Coxeter groups and unipotent representations. Number 212, pages 191–203. 1993. Représentations unipotentes génériques et blocs des groupes réductifs finis. URL: http://www.numdam.org/item/AST_1993__212__191_0/.

[Lus94] George Lusztig. Exotic Fourier transform. *Duke Math. J.*, 73(1):227–241, 243–248, 1994. With an appendix by Gunter Malle. doi:10.1215/S0012-7094-94-07309-2.

[Lus97] G. Lusztig. Notes on unipotent classes. *Asian J. Math.*, 1(1):194–207, 1997. doi:10.4310/AJM.1997.v1.n1.a7.

[Lus04] G. Lusztig. An induction theorem for Springer's representations. In *Representation theory of algebraic groups and quantum groups. Papers from the conference held as the 10th International Research Institute of the Mathematical Society of Japan (MSJ-IRI) at Sophia University, Tokyo, Japan, August 1–10, 2001*, pages 253–259. Tokyo: Mathematical Society of Japan, 2004.

[Lus11a] G. Lusztig. From conjugacy classes in the Weyl group to unipotent classes. *Represent. Theory*, 15:494–530, 2011. doi:10.1090/S1088-4165-2011-00396-4.

[Lus11b] George Lusztig. On some partitions of a flag manifold. *Asian J. Math.*, 15(1):1–8, 2011. doi:10.4310/AJM.2011.v15.n1.a1.

[Lus15] G. Lusztig. Truncated convolution of character sheaves. *Bull. Inst. Math. Acad. Sin. (N.S.)*, 10(1):1–72, 2015. URL: https://www.math.sinica.edu.tw/bulletin/journals/1613.

[Mac95] Ian Grant Macdonald. *Symmetric functions and Hall polynomials.* Oxford: Clarendon Press, 2nd edition, 1995.

[Mic23] Jean Michel. Tower equivalence and Lusztig's truncated Fourier transform. *Proc. Am. Math. Soc., Ser. B*, 10:252–261, 2023. doi:10.1090/bproc/167.

[Rea07] Nathan Reading. Clusters, Coxeter-sortable elements and noncrossing partitions. *Trans. Am. Math. Soc.*, 359(12):5931–5958, 2007. doi:10.1090/S0002-9947-07-04319-X.

[Sho88] Toshiaki Shoji. Geometry of orbits and Springer correspondence. In *Orbites unipotentes et représentations. I. Groupes finis et algèbres de Hecke*, pages 61–140. Paris: Société Mathématique de France, 1988.

[Som11] Eric Sommers. Exterior powers of the reflection representation in Springer theory. *Transform. Groups*, 16(3):889–911, errata 913, 2011. doi:10.1007/s00031-011-9148-3.

[Spr76] T. A. Springer. Trigonometric sums, Green functions of finite groups and representations of Weyl groups. *Invent. Math.*, 36:173–207, 1976. doi:10.1007/BF01390009.

[Spr78] T. A. Springer. A construction of representations of Weyl groups. *Invent. Math.*, 44:279–293, 1978. doi:10.1007/BF01403165.

[Spr84] T. A. Springer. A purity result for fixed point varieties in flag manifolds. *J. Fac. Sci., Univ. Tokyo, Sect. I A*, 31:271–282, 1984.

[SS93] Richard P. Stanley and John R. Stembridge. On immanants of Jacobi-Trudi matrices and permutations with restricted position. *J. Comb. Theory, Ser. A*, 62(2):261–279, 1993. doi:10.1016/0097-3165(93)90048-D.

[Sta95] Richard P. Stanley. A symmetric function generalization of the chromatic polynomial of a graph. *Adv. Math.*, 111(1):166–194, 1995. doi:10.1006/aima.1995.1020.

[SW12] John Shareshian and Michelle L. Wachs. Chromatic quasisymmetric functions and Hessenberg varieties. In *Configuration spaces. Geometry, combinatorics and topology*, pages 433–460. Pisa: Edizioni della Normale, 2012.

[Tri21] Minh-Tâm Quang Trinh. From the Hecke category to the unipotent locus, 2021. v2. arXiv:2106.07444.

[Wan26] Xiao Griffin Wang. *Multiplicative Hitchin Fibrations and the Fundamental Lemma*. Princeton University Press, Princeton, 2026. ISBN: 9780691281254.

[WW11] Ben Webster and Geordie Williamson. The geometry of Markov traces. *Duke Math. J.*, 160(2):401–419, 2011. doi:10.1215/00127094-1444268.

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